

Final coursework - Data Exploration and Visualisation

STAT5228_GroupB:

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Overview

This report presents an analysis of the data with the aim of identifying key business insights related to sales performance, customer behaviour, and operational efficiency. The dataset was first cleaned and validated to ensure accuracy, after which exploratory and focused analyses were conducted.

A specific focus was placed on the Moka district and the Electronics product category to enable deeper insights into high-value transactions and operational patterns.

Task 1 : Data audit, tidy data and cleaning

For Task 1, we have loaded the csv file in R software and has conducted the data audit, data tidying and cleaning. It is saved as `cleaned_data.csv` . We then filtered by district and product category and saved the focus data as `Moka_data.csv`.

```
##changing working directory
setwd("C:/Users/meerj/OneDrive/Desktop/Data Exploration and Visualisation/Final Assignment")

##TASK1:
data<-read.csv("STAT5228_Portfolio_Data.csv",header=T,na.strings=c(" ", "NA"))
View(data)
attach(data)
str(data)
```

```
## 'data.frame': 24000 obs. of 32 variables:
## $ transaction_id : chr "T0016367" "T0011807" "T0005406" "T0007504" ...
## $ datetime : chr "2024-01-01 00:33:41" "2024-01-01 04:20:26" "2024-01-01 04:27:32" "2024-01-01 04:27:32" ...
## $ date : chr "2024-01-01" "2024-01-01" "2024-01-01" "2024-01-01" ...
## $ year : int 2024 2024 2024 2024 2024 2024 2024 2024 2024 2024 ...
## $ month : int 1 1 1 1 1 1 1 1 1 1 ...
## $ year_month : chr "2024-01" "2024-01" "2024-01" "2024-01" ...
## $ year_quarter : chr "2024Q1" "2024Q1" "2024Q1" "2024Q1" ...
## $ store_id : int 12 5 3 1 2 5 10 8 12 8 ...
## $ store_name : chr "IslandMart Store 12" "IslandMart Store 05" "IslandMart Store 03" "IslandMart Store 03" ...
## $ district : chr "Plaines Wilhems" "Black River" "Plaines Wilhems" "Pamplemousses" ...
## $ store_lat : num -20.3 -20.3 -20.3 -20.1 -20.2 ...
## $ store_lon : num 57.5 57.4 57.5 57.6 57.5 ...
## $ customer_id : int 1574 1245 650 2156 653 589 2151 1848 40 35 ...
## $ customer_gender : chr "Male" "Male" "Female" "Male" ...
## $ age_band : chr "25-34" "25-34" "35-44" "18-24" ...
## $ customer_segment : chr "New" "New" "New" "Returning" ...
## $ channel : chr "In-store" "In-store" "In-store" "Delivery" ...
## $ payment_method : chr "Cash" "MobileMoney" "Card" "Cash" ...
## $ campaign : chr "None" "None" "None" "None" ...
## $ product_category : chr "Grocery" "Baby" "Grocery" "Grocery" ...
## $ product_subcategory : chr "Beverages" "Care" "Fresh" "Fresh" ...
## $ quantity : int 3 2 3 2 1 1 1 3 1 2 ...
## $ unit_price_mur : num 46.4 428.9 87.1 84.5 40.6 ...
## $ discount_pct : num 0.037 0.012 0.01 0.016 0.042 0.01 0.012 0.048 0.005 0.016 ...
## $ gross_margin_pct : num 0.192 0.18 0.184 0.204 0.273 0.237 0.243 0.231 0.268 0.298 ...
## $ gross_sales_mur : num 139.1 857.7 261.3 168.9 40.6 ...
## $ net_sales_mur : num 133.9 847.4 258.7 166.2 38.9 ...
## $ cost_mur : num 108.2 694.9 211.1 132.3 28.3 ...
```

```
## $ gross_profit_mur : num 25.7 152.5 47.6 33.9 10.6 ...
## $ returned_flag : int 0 0 0 0 0 0 0 0 0 0 ...
## $ delivery_minutes : num NA NA NA 72 NA 16 24 NA NA 14 ...
## $ satisfaction_score : num 4.6 4.5 3.6 4 4.4 5 NA 4.8 4.7 3.9 ...
```

```
summary(data)
```

```
## transaction_id      datetime      date      year
## Length:24000      Length:24000      Length:24000      Min. :2024
## Class :character      Class :character      Class :character      1st Qu.:2024
## Mode :character      Mode :character      Mode :character      Median :2024
##                                     Mean :2024
##                                     3rd Qu.:2025
##                                     Max. :2025
##
##      month      year_month      year_quarter      store_id
## Min. : 1.000      Length:24000      Length:24000      Min. : 1.000
## 1st Qu.: 4.000      Class :character      Class :character      1st Qu.: 4.000
## Median : 7.000      Mode :character      Mode :character      Median : 7.000
## Mean : 6.544                                     Mean : 6.552
## 3rd Qu.:10.000                                     3rd Qu.:10.000
## Max. :12.000                                     Max. :12.000
##
##      store_name      district      store_lat      store_lon
## Length:24000      Length:24000      Min. : -20.48      Min. :57.36
## Class :character      Class :character      1st Qu.: -20.38      1st Qu.:57.39
## Mode :character      Mode :character      Median : -20.28      Median :57.48
##                                     Mean : -20.26      Mean :57.49
##                                     3rd Qu.: -20.16      3rd Qu.:57.53
##                                     Max. : -20.04      Max. :57.72
##                                     NA's :144      NA's :144
##
##      customer_id      customer_gender      age_band      customer_segment
## Min. : 1      Length:24000      Length:24000      Length:24000
## 1st Qu.: 617      Class :character      Class :character      Class :character
## Median :1239      Mode :character      Mode :character      Mode :character
## Mean :1244
## 3rd Qu.:1868
## Max. :2500
##
##      channel      payment_method      campaign      product_category
## Length:24000      Length:24000      Length:24000      Length:24000
## Class :character      Class :character      Class :character      Class :character
## Mode :character      Mode :character      Mode :character      Mode :character
##
##
##
##      product_subcategory      quantity      unit_price_mur      discount_pct
## Length:24000      Min. : 0.000      Min. : -7243.96      Min. :0.00000
## Class :character      1st Qu.: 1.000      1st Qu.: 86.86      1st Qu.:0.04100
## Mode :character      Median : 2.000      Median : 257.14      Median :0.08200
##                                     Mean : 1.858      Mean : 871.89      Mean :0.09282
##                                     3rd Qu.: 2.000      3rd Qu.: 782.08      3rd Qu.:0.13900
##                                     Max. :76.000      Max. :15221.18      Max. :0.27900
```

```
##
## gross_margin_pct gross_sales_mur net_sales_mur cost_mur NA's :480
## Min. :0.0550 Min. : -7244.0 Min. : -7215.0 Min. : -5916.3
## 1st Qu.:0.2000 1st Qu.: 181.7 1st Qu.: 164.6 1st Qu.: 126.1
## Median :0.2310 Median : 417.7 Median : 376.6 Median : 282.2
## Mean :0.2379 Mean : 1125.4 Mean : 1022.1 Mean : 801.9
## 3rd Qu.:0.2710 3rd Qu.: 1160.3 3rd Qu.: 1053.8 3rd Qu.: 763.7
## Max. :0.4640 Max. :158662.4 Max. :125026.0 Max. :106522.1
##
## gross_profit_mur returned_flag delivery_minutes satisfaction_score
## Min. : -1298.70 Min. :0.00000 Min. : 5.00 Min. :2.000
## 1st Qu.: 37.45 1st Qu.:0.00000 1st Qu.: 22.00 1st Qu.:3.700
## Median : 90.94 Median :0.00000 Median : 33.00 Median :4.100
## Mean : 220.14 Mean :0.02854 Mean : 37.16 Mean :4.077
## 3rd Qu.: 278.30 3rd Qu.:0.00000 3rd Qu.: 51.00 3rd Qu.:4.400
## Max. :18503.85 Max. :1.00000 Max. :116.00 Max. :5.000
##
## NA's :16845 NA's :2400
```

```
###checking for any missingness of values
colSums(is.na(data)) %>% sort(decreasing = TRUE)
```

```
## delivery_minutes satisfaction_score discount_pct store_lat
## 16845 2400 480 144
## store_lon transaction_id datetime date
## 144 0 0 0
## year month year_month year_quarter
## 0 0 0 0
## store_id store_name district customer_id
## 0 0 0 0
## customer_gender age_band customer_segment channel
## 0 0 0 0
## payment_method campaign product_category product_subcategory
## 0 0 0 0
## quantity unit_price_mur gross_margin_pct gross_sales_mur
## 0 0 0 0
## net_sales_mur cost_mur gross_profit_mur returned_flag
## 0 0 0 0
```

```
###Variable 1: Transaction Id - checking for any duplications
```

```
duplicates <- data %>%
  count(transaction_id) %>%
  filter(n > 1)
```

```
nrow(duplicates)
```

```
## [1] 50
```

```
str(data)
```

```
## 'data.frame': 24000 obs. of 32 variables:
## $ transaction_id : chr "T0016367" "T0011807" "T0005406" "T0007504" ...
## $ datetime : chr "2024-01-01 00:33:41" "2024-01-01 04:20:26" "2024-01-01 04:27:32" "2024-01-01 04:27:32" ...
```

```
## $ date           : chr "2024-01-01" "2024-01-01" "2024-01-01" "2024-01-01" ...
## $ year           : int 2024 2024 2024 2024 2024 2024 2024 2024 2024 2024 ...
## $ month          : int 1 1 1 1 1 1 1 1 1 1 ...
## $ year_month     : chr "2024-01" "2024-01" "2024-01" "2024-01" ...
## $ year_quarter   : chr "2024Q1" "2024Q1" "2024Q1" "2024Q1" ...
## $ store_id       : int 12 5 3 1 2 5 10 8 12 8 ...
## $ store_name     : chr "IslandMart Store 12" "IslandMart Store 05" "IslandMart Store 03" "IslandMart Store 04" ...
## $ district       : chr "Plaines Wilhems" "Black River" "Plaines Wilhems" "Pamplemousses" ...
## $ store_lat      : num -20.3 -20.3 -20.3 -20.1 -20.2 ...
## $ store_lon      : num 57.5 57.4 57.5 57.6 57.5 ...
## $ customer_id    : int 1574 1245 650 2156 653 589 2151 1848 40 35 ...
## $ customer_gender : chr "Male" "Male" "Female" "Male" ...
## $ age_band       : chr "25-34" "25-34" "35-44" "18-24" ...
## $ customer_segment : chr "New" "New" "New" "Returning" ...
## $ channel        : chr "In-store" "In-store" "In-store" "Delivery" ...
## $ payment_method  : chr "Cash" "MobileMoney" "Card" "Cash" ...
## $ campaign       : chr "None" "None" "None" "None" ...
## $ product_category : chr "Grocery" "Baby" "Grocery" "Grocery" ...
## $ product_subcategory : chr "Beverages" "Care" "Fresh" "Fresh" ...
## $ quantity       : int 3 2 3 2 1 1 1 3 1 2 ...
## $ unit_price_mur  : num 46.4 428.9 87.1 84.5 40.6 ...
## $ discount_pct    : num 0.037 0.012 0.01 0.016 0.042 0.01 0.012 0.048 0.005 0.016 ...
## $ gross_margin_pct : num 0.192 0.18 0.184 0.204 0.273 0.237 0.243 0.231 0.268 0.298 ...
## $ gross_sales_mur : num 139.1 857.7 261.3 168.9 40.6 ...
## $ net_sales_mur   : num 133.9 847.4 258.7 166.2 38.9 ...
## $ cost_mur        : num 108.2 694.9 211.1 132.3 28.3 ...
## $ gross_profit_mur : num 25.7 152.5 47.6 33.9 10.6 ...
## $ returned_flag   : int 0 0 0 0 0 0 0 0 0 0 ...
## $ delivery_minutes : num NA NA NA 72 NA 16 24 NA NA 14 ...
## $ satisfaction_score : num 4.6 4.5 3.6 4 4.4 5 NA 4.8 4.7 3.9 ...
```

#from the above generated code, we have 50 duplications of transactions id

##removing of duplications

```
data <- data %>%
  distinct(transaction_id, .keep_all = TRUE)
```

#From the first summary we had 24000 observations for transaction ids

#After removing the duplications, we are left with 23950 - successfully remove the 50 duplications

####Variable 2: discount_pct

#we have 480 missing dicount percentage

##missing discount to be set to 0

```
data <- data %>%
  mutate(discount_pct = replace_na(discount_pct, 0))
```

##verifying dicount limit 0-0.35

```
data %>% filter(discount_pct < 0 | discount_pct > 0.35)
```

```
## [1] transaction_id    datetime           date
## [4] year              month             year_month
## [7] year_quarter      store_id          store_name
## [10] district          store_lat         store_lon
```

```
## [13] customer_id      customer_gender    age_band
## [16] customer_segment  channel            payment_method
## [19] campaign          product_category   product_subcategory
## [22] quantity          unit_price_mur     discount_pct
## [25] gross_margin_pct  gross_sales_mur    net_sales_mur
## [28] cost_mur          gross_profit_mur   returned_flag
## [31] delivery_minutes  satisfaction_score
## <0 rows> (or 0-length row.names)
```

```
data <- data %>% filter(discount_pct >= 0, discount_pct <= 0.35)
```

```
###Variable 3 : quantity, unit price MUR
```

```
data %>% filter(quantity <= 0 | unit_price_mur <= 0)
```

##	transaction_id	datetime	date	year	month	year_month
## 1	T0023342	2024-01-08 11:18:24	2024-01-08	2024	1	2024-01
## 2	T0010349	2024-02-22 12:24:44	2024-02-22	2024	2	2024-02
## 3	T0009287	2024-03-07 03:47:16	2024-03-07	2024	3	2024-03
## 4	T0010226	2024-03-16 19:50:24	2024-03-16	2024	3	2024-03
## 5	T0016373	2024-04-09 07:34:45	2024-04-09	2024	4	2024-04
## 6	T0021938	2024-04-14 02:52:18	2024-04-14	2024	4	2024-04
## 7	T0000322	2024-04-15 18:26:24	2024-04-15	2024	4	2024-04
## 8	T0014793	2024-05-03 11:06:32	2024-05-03	2024	5	2024-05
## 9	T0003832	2024-05-07 15:08:03	2024-05-07	2024	5	2024-05
## 10	T0001383	2024-05-12 18:08:57	2024-05-12	2024	5	2024-05
## 11	T0008196	2024-05-22 09:11:34	2024-05-22	2024	5	2024-05
## 12	T0019709	2024-06-08 13:24:56	2024-06-08	2024	6	2024-06
## 13	T0008051	2024-06-10 15:07:13	2024-06-10	2024	6	2024-06
## 14	T0010582	2024-06-19 09:34:57	2024-06-19	2024	6	2024-06
## 15	T0013127	2024-06-21 13:44:37	2024-06-21	2024	6	2024-06
## 16	T0017478	2024-07-20 05:41:03	2024-07-20	2024	7	2024-07
## 17	T0013950	2024-08-04 03:04:57	2024-08-04	2024	8	2024-08
## 18	T0002396	2024-08-04 20:44:10	2024-08-04	2024	8	2024-08
## 19	T0008082	2024-08-17 13:43:05	2024-08-17	2024	8	2024-08
## 20	T0020786	2024-09-10 11:47:15	2024-09-10	2024	9	2024-09
## 21	T0019959	2024-10-08 03:28:00	2024-10-08	2024	10	2024-10
## 22	T0008742	2024-10-25 07:49:00	2024-10-25	2024	10	2024-10
## 23	T0002403	2024-11-16 20:52:20	2024-11-16	2024	11	2024-11
## 24	T0002881	2024-11-19 14:33:38	2024-11-19	2024	11	2024-11
## 25	T0006028	2024-11-20 21:26:16	2024-11-20	2024	11	2024-11
## 26	T0013425	2024-11-25 22:18:44	2024-11-25	2024	11	2024-11
## 27	T0003004	2024-12-13 08:50:24	2024-12-13	2024	12	2024-12
## 28	T0019402	2024-12-31 06:18:35	2024-12-31	2024	12	2024-12
## 29	T0023390	2025-01-14 15:26:34	2025-01-14	2025	1	2025-01
## 30	T0003326	2025-01-22 03:30:20	2025-01-22	2025	1	2025-01
## 31	T0013348	2025-01-29 08:35:07	2025-01-29	2025	1	2025-01
## 32	T0003237	2025-02-20 19:26:17	2025-02-20	2025	2	2025-02
## 33	T0013701	2025-03-04 06:55:50	2025-03-04	2025	3	2025-03
## 34	T0017552	2025-03-14 16:56:03	2025-03-14	2025	3	2025-03
## 35	T0005795	2025-03-28 14:24:23	2025-03-28	2025	3	2025-03
## 36	T0022132	2025-03-30 10:36:08	2025-03-30	2025	3	2025-03
## 37	T0008062	2025-04-15 19:22:01	2025-04-15	2025	4	2025-04
## 38	T0006966	2025-04-20 13:01:45	2025-04-20	2025	4	2025-04
## 39	T0000462	2025-04-28 11:34:30	2025-04-28	2025	4	2025-04

## 40	T0018771	2025-05-06	04:16:39	2025-05-06	2025	5	2025-05
## 41	T0006173	2025-05-09	16:53:54	2025-05-09	2025	5	2025-05
## 42	T0000219	2025-06-18	04:51:03	2025-06-18	2025	6	2025-06
## 43	T0011533	2025-06-26	13:42:10	2025-06-26	2025	6	2025-06
## 44	T0006992	2025-07-17	10:03:07	2025-07-17	2025	7	2025-07
## 45	T0009618	2025-08-02	01:48:20	2025-08-02	2025	8	2025-08
## 46	T0022402	2025-08-04	06:01:13	2025-08-04	2025	8	2025-08
## 47	T0013719	2025-08-25	15:35:58	2025-08-25	2025	8	2025-08
## 48	T0016100	2025-08-29	18:18:22	2025-08-29	2025	8	2025-08
## 49	T0020131	2025-09-13	09:03:01	2025-09-13	2025	9	2025-09
## 50	T0014413	2025-09-22	08:57:44	2025-09-22	2025	9	2025-09
## 51	T0022937	2025-09-25	13:22:21	2025-09-25	2025	9	2025-09
## 52	T0006863	2025-10-01	21:41:24	2025-10-01	2025	10	2025-10
## 53	T0019590	2025-10-12	20:50:59	2025-10-12	2025	10	2025-10
## 54	T0001884	2025-10-16	15:06:17	2025-10-16	2025	10	2025-10
## 55	T0005627	2025-10-19	14:54:36	2025-10-19	2025	10	2025-10
## 56	T0018287	2025-10-23	11:21:54	2025-10-23	2025	10	2025-10
## 57	T0021635	2025-10-25	02:00:18	2025-10-25	2025	10	2025-10
## 58	T0013544	2025-11-24	05:09:37	2025-11-24	2025	11	2025-11
## 59	T0019695	2025-12-25	02:53:10	2025-12-25	2025	12	2025-12
## 60	T0012113	2025-12-27	14:55:30	2025-12-27	2025	12	2025-12
##	year_quarter	store_id	store_name	district	store_lat		
## 1	2024Q1	4	IslandMart Store 04	Pamplemousses	-20.09746		
## 2	2024Q1	1	IslandMart Store 01	Pamplemousses	-20.12781		
## 3	2024Q1	6	IslandMart Store 06	Rivière du Rempart	-20.03654		
## 4	2024Q1	8	IslandMart Store 08	Savanne	-20.48387		
## 5	2024Q2	8	IslandMart Store 08	Savanne	-20.48387		
## 6	2024Q2	2	IslandMart Store 02	Moka	-20.19017		
## 7	2024Q2	10	IslandMart Store 10	Black River	-20.39135		
## 8	2024Q2	4	IslandMart Store 04	Pamplemousses	-20.09746		
## 9	2024Q2	6	IslandMart Store 06	Rivière du Rempart	-20.03654		
## 10	2024Q2	7	IslandMart Store 07	Black River	-20.37842		
## 11	2024Q2	11	IslandMart Store 11	Moka	-20.21830		
## 12	2024Q2	11	IslandMart Store 11	Moka	-20.21830		
## 13	2024Q2	11	IslandMart Store 11	Moka	-20.21830		
## 14	2024Q2	10	IslandMart Store 10	Black River	NA		
## 15	2024Q2	7	IslandMart Store 07	Black River	-20.37842		
## 16	2024Q3	12	IslandMart Store 12	Plaines Wilhems	-20.28937		
## 17	2024Q3	6	IslandMart Store 06	Rivière du Rempart	-20.03654		
## 18	2024Q3	11	IslandMart Store 11	Moka	-20.21830		
## 19	2024Q3	11	IslandMart Store 11	Moka	-20.21830		
## 20	2024Q3	2	IslandMart Store 02	Moka	-20.19017		
## 21	2024Q4	4	IslandMart Store 04	Pamplemousses	-20.09746		
## 22	2024Q4	2	IslandMart Store 02	Moka	-20.19017		
## 23	2024Q4	6	IslandMart Store 06	Rivière du Rempart	-20.03654		
## 24	2024Q4	8	IslandMart Store 08	Savanne	-20.48387		
## 25	2024Q4	2	IslandMart Store 02	Moka	-20.19017		
## 26	2024Q4	2	IslandMart Store 02	Moka	-20.19017		
## 27	2024Q4	6	IslandMart Store 06	Rivière du Rempart	-20.03654		
## 28	2024Q4	2	IslandMart Store 02	Moka	-20.19017		
## 29	2025Q1	9	IslandMart Store 09	Port Louis	-20.15831		
## 30	2025Q1	9	IslandMart Store 09	Port Louis	-20.15831		
## 31	2025Q1	8	IslandMart Store 08	Savanne	-20.48387		
## 32	2025Q1	4	IslandMart Store 04	Pamplemousses	-20.09746		

## 33	2025Q1	12	IslandMart	Store 12	Plaines Wilhems	-20.28937
## 34	2025Q1	2	IslandMart	Store 02	Moka	-20.19017
## 35	2025Q1	5	IslandMart	Store 05	Black River	-20.34486
## 36	2025Q1	11	IslandMart	Store 11	Moka	-20.21830
## 37	2025Q2	10	IslandMart	Store 10	Black River	-20.39135
## 38	2025Q2	7	IslandMart	Store 07	Black River	-20.37842
## 39	2025Q2	6	IslandMart	Store 06	Rivière du Rempart	-20.03654
## 40	2025Q2	3	IslandMart	Store 03	Plaines Wilhems	-20.28385
## 41	2025Q2	10	IslandMart	Store 10	Black River	-20.39135
## 42	2025Q2	8	IslandMart	Store 08	Savanne	-20.48387
## 43	2025Q2	3	IslandMart	Store 03	Plaines Wilhems	-20.28385
## 44	2025Q3	2	IslandMart	Store 02	Moka	-20.19017
## 45	2025Q3	3	IslandMart	Store 03	Plaines Wilhems	-20.28385
## 46	2025Q3	5	IslandMart	Store 05	Black River	-20.34486
## 47	2025Q3	11	IslandMart	Store 11	Moka	-20.21830
## 48	2025Q3	11	IslandMart	Store 11	Moka	-20.21830
## 49	2025Q3	8	IslandMart	Store 08	Savanne	-20.48387
## 50	2025Q3	12	IslandMart	Store 12	Plaines Wilhems	-20.28937
## 51	2025Q3	6	IslandMart	Store 06	Rivière du Rempart	-20.03654
## 52	2025Q4	10	IslandMart	Store 10	Black River	-20.39135
## 53	2025Q4	9	IslandMart	Store 09	Port Louis	-20.15831
## 54	2025Q4	10	IslandMart	Store 10	Black River	-20.39135
## 55	2025Q4	10	IslandMart	Store 10	Black River	-20.39135
## 56	2025Q4	11	IslandMart	Store 11	Moka	-20.21830
## 57	2025Q4	2	IslandMart	Store 02	Moka	-20.19017
## 58	2025Q4	3	IslandMart	Store 03	Plaines Wilhems	-20.28385
## 59	2025Q4	10	IslandMart	Store 10	Black River	-20.39135
## 60	2025Q4	1	IslandMart	Store 01	Pamplemousses	-20.12781
##	store_lon	customer_id	customer_gender	age_band	customer_segment	channel
## 1	57.57448	1430	Male	55+	Returning	In-store
## 2	57.56431	716	Female	55+	Returning	In-store
## 3	57.72495	2099	Male	55+	VIP	Delivery
## 4	57.53127	2411	Female	35-44	Returning	In-store
## 5	57.53127	2493	Female	18-24	Returning	Online
## 6	57.47793	1687	Male	18-24	Returning	In-store
## 7	57.39402	1044	Female	25-34	New	In-store
## 8	57.57448	92	Male	25-34	New	In-store
## 9	57.72495	1957	Male	55+	New	In-store
## 10	57.37749	1036	Female	25-34	Returning	Delivery
## 11	57.48904	1992	Female	18-24	Returning	In-store
## 12	57.48904	1993	Male	25-34	Returning	In-store
## 13	57.48904	86	Female	18-24	New	In-store
## 14	NA	80	Female	55+	Returning	In-store
## 15	57.37749	220	Female	45-54	New	In-store
## 16	57.45593	1984	Male	18-24	New	In-store
## 17	57.72495	985	Male	25-34	Returning	In-store
## 18	57.48904	2022	Female	45-54	Returning	Delivery
## 19	57.48904	503	Male	25-34	Returning	In-store
## 20	57.47793	1533	Female	18-24	Returning	In-store
## 21	57.57448	2277	Male	55+	VIP	In-store
## 22	57.47793	1728	PreferNotSay	25-34	New	Online
## 23	57.72495	1616	Female	25-34	Returning	In-store
## 24	57.53127	2234	Female	25-34	Returning	Online
## 25	57.47793	1476	Male	55+	Returning	In-store

## 26	57.47793	1814	Female	45-54	VIP In-store
## 27	57.72495	234	Female	35-44	VIP In-store
## 28	57.47793	2380	Female	45-54	Returning In-store
## 29	57.47251	2268	Female	18-24	Returning In-store
## 30	57.47251	138	Female	25-34	Returning In-store
## 31	57.53127	460	PreferNotSay	55+	Returning In-store
## 32	57.57448	1822	Female	18-24	Returning Online
## 33	57.45593	1139	Male	35-44	New Online
## 34	57.47793	1295	Male	45-54	New In-store
## 35	57.36261	1699	Male	45-54	New Online
## 36	57.48904	1894	Female	35-44	Returning In-store
## 37	57.39402	1740	Female	55+	Returning In-store
## 38	57.37749	977	Female	35-44	Returning In-store
## 39	57.72495	2140	Male	25-34	VIP Online
## 40	57.49057	1094	Female	25-34	New Online
## 41	57.39402	207	Female	25-34	Returning Online
## 42	57.53127	1438	Female	25-34	New In-store
## 43	57.49057	1997	Female	35-44	Returning In-store
## 44	57.47793	1936	Male	25-34	New In-store
## 45	57.49057	2266	Male	25-34	New In-store
## 46	57.36261	2446	Female	45-54	Returning In-store
## 47	57.48904	1632	Female	18-24	Returning Online
## 48	57.48904	975	Female	25-34	New In-store
## 49	57.53127	2312	Female	35-44	New Online
## 50	57.45593	1287	Male	45-54	New In-store
## 51	57.72495	1198	Male	18-24	Returning In-store
## 52	57.39402	1660	Female	25-34	Returning In-store
## 53	57.47251	1956	Male	25-34	New In-store
## 54	57.39402	1309	Female	18-24	Returning Delivery
## 55	57.39402	1380	Female	35-44	Returning Online
## 56	57.48904	2489	Male	25-34	New Online
## 57	57.47793	409	Male	25-34	New In-store
## 58	57.49057	2314	Male	25-34	New In-store
## 59	57.39402	731	Male	35-44	New In-store
## 60	57.56431	2339	Female	25-34	Returning In-store
##	payment_method	campaign	product_category	product_subcategory	quantity
## 1	Card	None	Baby	Diapers	3
## 2	Card	None	Pharmacy	OTC	1
## 3	MobileMoney	RamadanPromo	Pharmacy	Wellness	1
## 4	Cash	RamadanPromo	Pharmacy	OTC	1
## 5	Card	RamadanPromo	Beauty	Fragrance	0
## 6	Cash	RamadanPromo	Grocery	Fresh	0
## 7	Card	RamadanPromo	Home	Cleaning	2
## 8	Card	None	Grocery	Snacks	0
## 9	Cash	None	Beauty	Makeup	0
## 10	Cash	None	Baby	Formula	0
## 11	Card	None	Home	Kitchen	1
## 12	Cash	WinterSale	Grocery	Beverages	0
## 13	Cash	WinterSale	Electronics	Mobile	1
## 14	Card	WinterSale	Electronics	Accessories	1
## 15	Card	WinterSale	Grocery	Fresh	0
## 16	Cash	WinterSale	Fashion	Men	1
## 17	Card	BackToSchool	Beauty	Skincare	2
## 18	Card	BackToSchool	Home	Cleaning	1

## 19	Card	BackToSchool	Pharmacy	Wellness	1
## 20	Card	BackToSchool	Pharmacy	Wellness	1
## 21	Card	None	Electronics	Appliances	1
## 22	Card	None	Fashion	Men	0
## 23	Cash	FestiveDeals	Grocery	Fresh	3
## 24	MobileMoney	FestiveDeals	Fashion	Footwear	1
## 25	Cash	FestiveDeals	Baby	Diapers	0
## 26	Cash	FestiveDeals	Grocery	Snacks	1
## 27	Cash	FestiveDeals	Grocery	Beverages	3
## 28	Card	FestiveDeals	Electronics	Appliances	0
## 29	Card	None	Grocery	Dry	0
## 30	Card	None	Electronics	Accessories	1
## 31	Cash	None	Pharmacy	Wellness	2
## 32	Card	None	Fashion	Men	1
## 33	Card	RamadanPromo	Home	Cleaning	0
## 34	Cash	RamadanPromo	Grocery	Dry	2
## 35	Card	RamadanPromo	Pharmacy	Personal Care	0
## 36	Cash	RamadanPromo	Grocery	Snacks	2
## 37	Card	RamadanPromo	Grocery	Beverages	1
## 38	Card	RamadanPromo	Electronics	Appliances	1
## 39	Card	RamadanPromo	Pharmacy	OTC	1
## 40	Card	None	Pharmacy	OTC	0
## 41	BankTransfer	None	Baby	Care	0
## 42	Cash	WinterSale	Beauty	Fragrance	1
## 43	Card	WinterSale	Grocery	Fresh	0
## 44	MobileMoney	WinterSale	Fashion	Men	2
## 45	Card	BackToSchool	Grocery	Beverages	0
## 46	Card	BackToSchool	Pharmacy	OTC	1
## 47	MobileMoney	BackToSchool	Pharmacy	OTC	0
## 48	Cash	BackToSchool	Pharmacy	OTC	1
## 49	Card	BackToSchool	Fashion	Men	2
## 50	Cash	BackToSchool	Grocery	Dry	0
## 51	MobileMoney	BackToSchool	Fashion	Women	3
## 52	MobileMoney	None	Pharmacy	Wellness	0
## 53	Cash	None	Pharmacy	Wellness	1
## 54	BankTransfer	None	Grocery	Fresh	0
## 55	Card	None	Pharmacy	OTC	1
## 56	BankTransfer	None	Home	Decor	0
## 57	Card	None	Grocery	Beverages	4
## 58	Cash	FestiveDeals	Grocery	Fresh	0
## 59	Cash	FestiveDeals	Pharmacy	Personal Care	1
## 60	Cash	FestiveDeals	Beauty	Skincare	0
##	unit_price_mur	discount_pct	gross_margin_pct	gross_sales_mur	net_sales_mur
## 1	-1039.09	0.017	0.214	-3117.27	-3064.28
## 2	-249.04	0.018	0.264	-249.04	-244.56
## 3	-296.23	0.114	0.249	-296.23	-262.46
## 4	-300.00	0.034	0.289	-300.00	-289.80
## 5	703.59	0.101	0.298	0.00	0.00
## 6	69.28	0.125	0.196	0.00	0.00
## 7	-418.50	0.143	0.199	-837.00	-717.31
## 8	43.30	0.012	0.186	0.00	0.00
## 9	1101.35	0.019	0.334	0.00	0.00
## 10	856.69	0.043	0.210	0.00	0.00
## 11	-527.00	0.027	0.195	-527.00	-512.77

## 12	126.77	0.145	0.192	0.00	0.00
## 13	-4406.63	0.081	0.193	-4406.63	-4049.69
## 14	-4106.37	0.105	0.119	-4106.37	-3675.20
## 15	114.30	0.200	0.246	0.00	0.00
## 16	-1080.53	0.113	0.396	-1080.53	-958.43
## 17	-607.14	0.081	0.291	-1214.28	-1115.92
## 18	-535.16	0.037	0.181	-535.16	-515.36
## 19	-199.57	0.104	0.241	-199.57	-178.81
## 20	-223.40	0.059	0.336	-223.40	-210.22
## 21	-3194.86	0.046	0.164	-3194.86	-3047.90
## 22	1070.27	0.015	0.354	0.00	0.00
## 23	-58.77	0.080	0.219	-176.31	-162.21
## 24	-798.60	0.245	0.299	-798.60	-602.94
## 25	453.35	0.187	0.141	0.00	0.00
## 26	-41.30	0.230	0.287	-41.30	-31.80
## 27	-79.24	0.084	0.284	-237.72	-217.75
## 28	2884.11	0.118	0.156	0.00	0.00
## 29	70.10	0.021	0.206	0.00	0.00
## 30	-7243.96	0.004	0.180	-7243.96	-7214.98
## 31	-217.86	0.022	0.310	-435.72	-426.13
## 32	-1192.02	0.012	0.322	-1192.02	-1177.72
## 33	627.47	0.058	0.160	0.00	0.00
## 34	-26.85	0.048	0.208	-53.70	-51.12
## 35	153.59	0.074	0.275	0.00	0.00
## 36	-108.23	0.033	0.248	-216.46	-209.32
## 37	-93.19	0.145	0.252	-93.19	-79.68
## 38	-4964.20	0.000	0.163	-4964.20	-4964.20
## 39	-280.29	0.143	0.265	-280.29	-240.21
## 40	246.79	0.018	0.286	0.00	0.00
## 41	400.09	0.027	0.213	0.00	0.00
## 42	-868.11	0.117	0.265	-868.11	-766.54
## 43	92.54	0.173	0.282	0.00	0.00
## 44	-768.67	0.121	0.368	-1537.34	-1351.32
## 45	112.43	0.125	0.214	0.00	0.00
## 46	-360.25	0.123	0.252	-360.25	-315.94
## 47	297.06	0.000	0.267	0.00	0.00
## 48	-270.15	0.092	0.295	-270.15	-245.30
## 49	-1352.96	0.163	0.357	-2705.92	-2264.86
## 50	71.14	0.087	0.172	0.00	0.00
## 51	-1230.07	0.045	0.354	-3690.21	-3524.15
## 52	306.45	0.011	0.328	0.00	0.00
## 53	-112.46	0.035	0.262	-112.46	-108.52
## 54	103.40	0.030	0.245	0.00	0.00
## 55	-90.00	0.042	0.316	-90.00	-86.22
## 56	505.44	0.016	0.212	0.00	0.00
## 57	-204.15	0.026	0.210	-816.60	-795.37
## 58	80.41	0.209	0.215	0.00	0.00
## 59	-275.33	0.084	0.285	-275.33	-252.20
## 60	604.10	0.197	0.269	0.00	0.00
##	cost_mur	gross_profit_mur	returned_flag	delivery_minutes	satisfaction_score
## 1	-2408.52	-655.76	0	NA	4.0
## 2	-180.00	-64.56	0	NA	4.5
## 3	-197.11	-65.35	0	78	3.1
## 4	-206.05	-83.75	0	NA	3.7

## 5	0.00	0.00	0	36	3.7
## 6	0.00	0.00	0	NA	4.5
## 7	-574.57	-142.74	0	NA	4.3
## 8	0.00	0.00	0	NA	4.3
## 9	0.00	0.00	0	NA	4.5
## 10	0.00	0.00	0	77	4.1
## 11	-412.78	-99.99	0	NA	NA
## 12	0.00	0.00	0	NA	3.9
## 13	-3268.10	-781.59	0	NA	4.3
## 14	-3237.85	-437.35	0	NA	4.4
## 15	0.00	0.00	0	NA	4.3
## 16	-578.89	-379.54	0	NA	NA
## 17	-791.19	-324.73	1	NA	3.1
## 18	-422.08	-93.28	0	35	3.6
## 19	-135.72	-43.09	0	NA	4.7
## 20	-139.59	-70.63	0	NA	4.9
## 21	-2548.04	-499.86	0	NA	NA
## 22	0.00	0.00	0	18	3.9
## 23	-126.69	-35.52	0	NA	3.3
## 24	-422.66	-180.28	0	7	4.4
## 25	0.00	0.00	0	NA	3.5
## 26	-22.67	-9.13	0	NA	3.8
## 27	-155.91	-61.84	0	NA	3.6
## 28	0.00	0.00	0	NA	3.9
## 29	0.00	0.00	0	NA	3.9
## 30	-5916.28	-1298.70	0	NA	3.8
## 31	-294.03	-132.10	0	NA	3.9
## 32	-798.49	-379.23	0	15	3.3
## 33	0.00	0.00	0	20	5.0
## 34	-40.49	-10.63	0	NA	4.5
## 35	0.00	0.00	1	5	2.4
## 36	-157.41	-51.91	0	NA	3.4
## 37	-59.60	-20.08	0	NA	4.1
## 38	-4155.04	-809.16	0	NA	3.5
## 39	-176.55	-63.66	0	18	3.5
## 40	0.00	0.00	0	37	3.8
## 41	0.00	0.00	0	5	4.0
## 42	-563.41	-203.13	0	NA	3.8
## 43	0.00	0.00	0	NA	NA
## 44	-854.03	-497.29	0	NA	4.3
## 45	0.00	0.00	0	NA	4.1
## 46	-236.32	-79.62	0	NA	4.1
## 47	0.00	0.00	0	9	4.5
## 48	-172.94	-72.36	0	NA	NA
## 49	-1456.30	-808.56	0	37	3.9
## 50	0.00	0.00	0	NA	NA
## 51	-2276.60	-1247.55	0	NA	4.1
## 52	0.00	0.00	0	NA	4.5
## 53	-80.09	-28.43	0	NA	4.2
## 54	0.00	0.00	0	58	4.7
## 55	-58.97	-27.25	1	37	3.8
## 56	0.00	0.00	1	39	4.6
## 57	-628.34	-167.03	0	NA	3.8
## 58	0.00	0.00	0	NA	4.5

```
## 59 -180.32      -71.88      0      NA      4.3
## 60   0.00       0.00      0      NA      4.0
```

```
data <- data %>% filter(quantity > 0, unit_price_mur > 0)

##removed invalid values

###variable 4: gross_sales_mur, net_sales_mur, cost_mur, gross_profit_mur

###addressing inconsistencies
##net sales = gross * (1-discount)

data <- data %>%
  mutate(calc_net_sales = gross_sales_mur * (1 - discount_pct),
         diff = abs(calc_net_sales - net_sales_mur))

summary(data$diff)
```

```
##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.000000 0.001180 0.002440 0.002442 0.003720 0.005000
```

```
##rounding off the small differences
data <- data %>%
  mutate(net_sales_mur = calc_net_sales)

summary(data)
```

```
## transaction_id      datetime      date      year
## Length:23890      Length:23890      Length:23890      Min. :2024
## Class :character      Class :character      Class :character      1st Qu.:2024
## Mode :character      Mode :character      Mode :character      Median :2024
##                                     Mean :2024
##                                     3rd Qu.:2025
##                                     Max. :2025
##
##      month      year_month      year_quarter      store_id
## Min. : 1.000      Length:23890      Length:23890      Min. : 1.000
## 1st Qu.: 4.000      Class :character      Class :character      1st Qu.: 4.000
## Median : 7.000      Mode :character      Mode :character      Median : 7.000
## Mean : 6.542                                     Mean : 6.553
## 3rd Qu.:10.000                                     3rd Qu.:10.000
## Max. :12.000                                     Max. :12.000
##
##      store_name      district      store_lat      store_lon
## Length:23890      Length:23890      Min. : -20.48      Min. :57.36
## Class :character      Class :character      1st Qu.: -20.38      1st Qu.:57.39
## Mode :character      Mode :character      Median : -20.28      Median :57.48
##                                     Mean : -20.26      Mean :57.49
##                                     3rd Qu.: -20.16      3rd Qu.:57.53
##                                     Max. : -20.04      Max. :57.72
##                                     NA's :141      NA's :141
##      customer_id      customer_gender      age_band      customer_segment
## Min. : 1      Length:23890      Length:23890      Length:23890
```

```
## 1st Qu.: 617   Class :character   Class :character   Class :character
## Median :1238   Mode  :character   Mode  :character   Mode  :character
## Mean    :1244
## 3rd Qu.:1868
## Max.    :2500
##
##      channel      payment_method      campaign      product_category
## Length:23890     Length:23890     Length:23890     Length:23890
## Class :character   Class :character   Class :character   Class :character
## Mode  :character   Mode  :character   Mode  :character   Mode  :character
##
##
##
## product_subcategory  quantity      unit_price_mur      discount_pct
## Length:23890        Min.   : 1.000      Min.   : 22.56      Min.   :0.000
## Class :character     1st Qu.: 1.000      1st Qu.: 87.01      1st Qu.:0.039
## Mode  :character     Median : 2.000      Median : 257.87     Median :0.080
##                      Mean   : 1.861      Mean   : 873.94     Mean   :0.091
##                      3rd Qu.: 2.000      3rd Qu.: 782.38     3rd Qu.:0.137
##                      Max.   :76.000      Max.   :15221.18     Max.   :0.279
##
## gross_margin_pct gross_sales_mur      net_sales_mur      cost_mur
## Min.   :0.0550   Min.   : 25.04      Min.   : 19.83      Min.   : 15.43
## 1st Qu.:0.2000   1st Qu.: 182.67     1st Qu.: 165.36     1st Qu.: 126.77
## Median :0.2310   Median : 419.85     Median : 378.15     Median : 283.68
## Mean   :0.2379   Mean   : 1129.15     Mean   : 1025.48     Mean   : 804.47
## 3rd Qu.:0.2710   3rd Qu.: 1162.38     3rd Qu.: 1055.94     3rd Qu.: 765.30
## Max.   :0.4640   Max.   :158662.44     Max.   :125026.00     Max.   :106522.15
##
## gross_profit_mur  returned_flag      delivery_minutes satisfaction_score
## Min.   : 4.05      Min.   :0.00000      Min.   : 5.00      Min.   :2.000
## 1st Qu.: 37.64     1st Qu.:0.00000      1st Qu.: 22.00     1st Qu.:3.700
## Median : 91.39     Median :0.00000      Median : 33.00     Median :4.100
## Mean   : 221.01     Mean   :0.02842      Mean   : 37.18     Mean   :4.077
## 3rd Qu.: 278.70     3rd Qu.:0.00000      3rd Qu.: 51.00     3rd Qu.:4.400
## Max.   :18503.85     Max.   :1.00000      Max.   :116.00     Max.   :5.000
##
##                      NA's :16768      NA's :2384
## calc_net_sales      diff
## Min.   : 19.83      Min.   :0.000000
## 1st Qu.: 165.36     1st Qu.:0.001180
## Median : 378.15     Median :0.002440
## Mean   : 1025.48     Mean   :0.002442
## 3rd Qu.: 1055.94     3rd Qu.:0.003720
## Max.   :125026.00     Max.   :0.005000
##
```

```
###variable 5: delivery times : in store
```

```
data <- data %>%
```

```
  mutate(delivery_minutes = ifelse(channel == "In-store", NA, delivery_minutes))
```

```
###variable 6: net sales mur
```

```
##checking for outliers
```

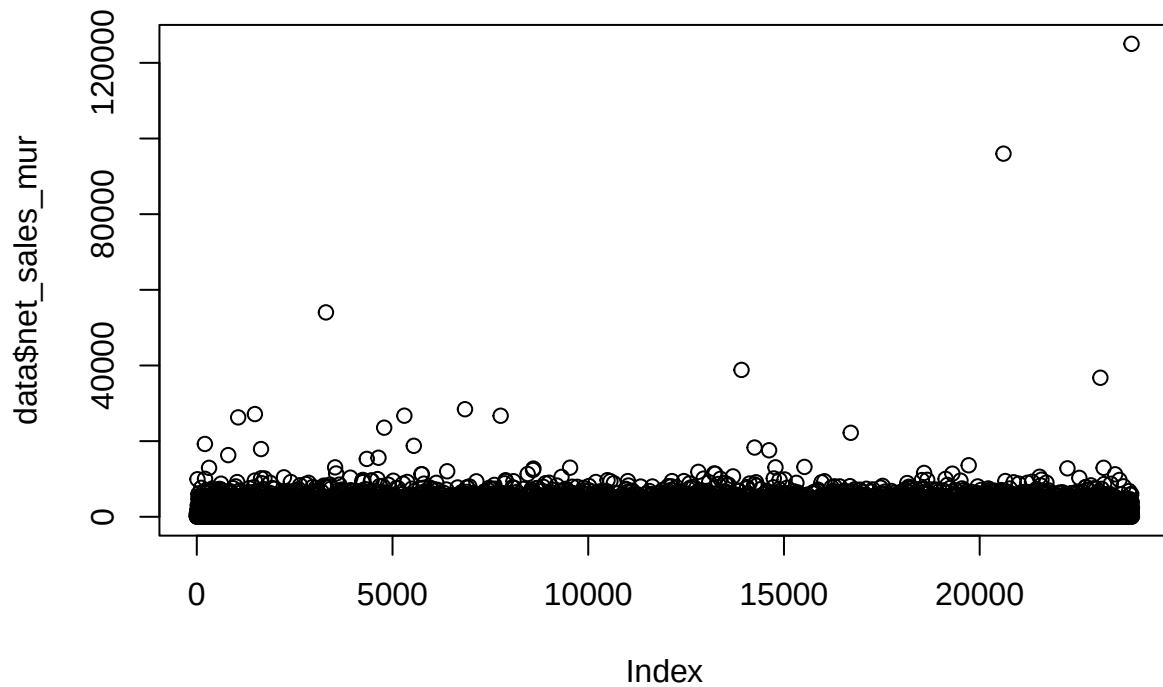
```
quantile(data$net_sales_mur, 0.99, na.rm = TRUE)
```

```
##      99%  
## 7034.871
```

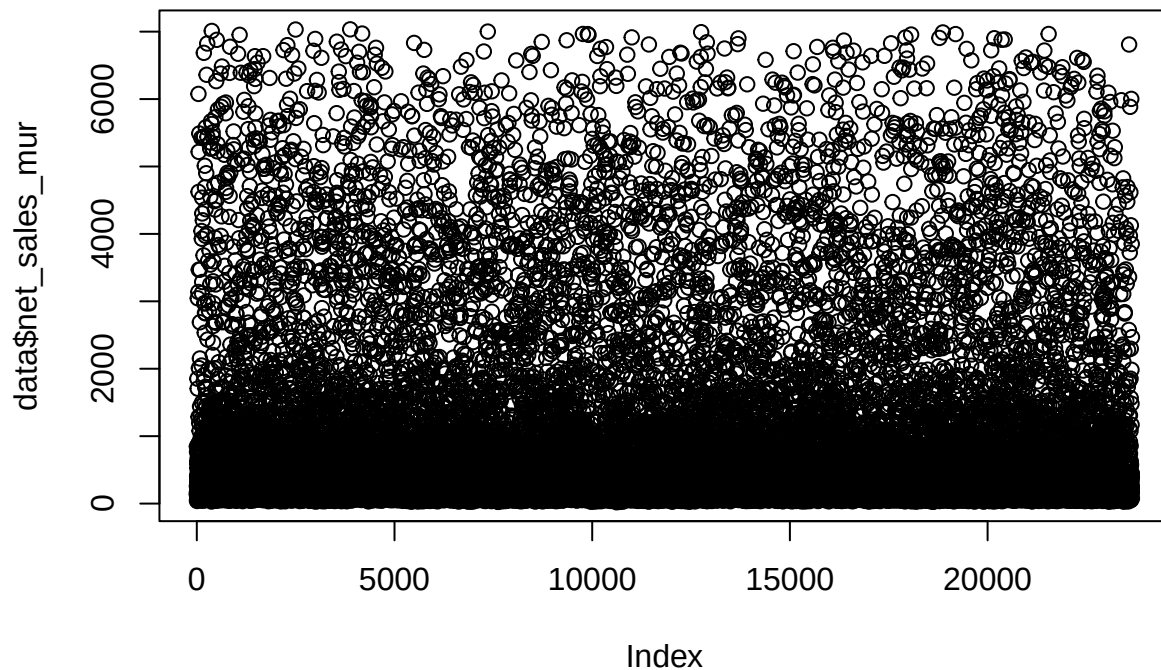
```
upper <- quantile(data$net_sales_mur, 0.99, na.rm = TRUE)  
upper
```

```
##      99%  
## 7034.871
```

```
plot(data$net_sales_mur)
```



```
data <- data %>% filter(net_sales_mur < upper)  
plot(data$net_sales_mur)
```



```
###removed the extra 1% outliers on top
```

```
###Variable 7 : date
```

```
data <- data %>%
  mutate(datetime = ymd_hms(datetime))
data <- data %>% filter(!is.na(datetime))
```

```
###variable 8 : satisfaction score
```

```
sum(is.na(data$satisfaction_score))
```

```
## [1] 2363
```

```
## we have 2384 missing values
```

```
mean(is.na(data$satisfaction_score)) * 100
```

```
## [1] 9.991121
```

```
## 9.991 % of the data is missing
```

```
##check where the missingness is coming from
##is it from delivery, in store or online??
```

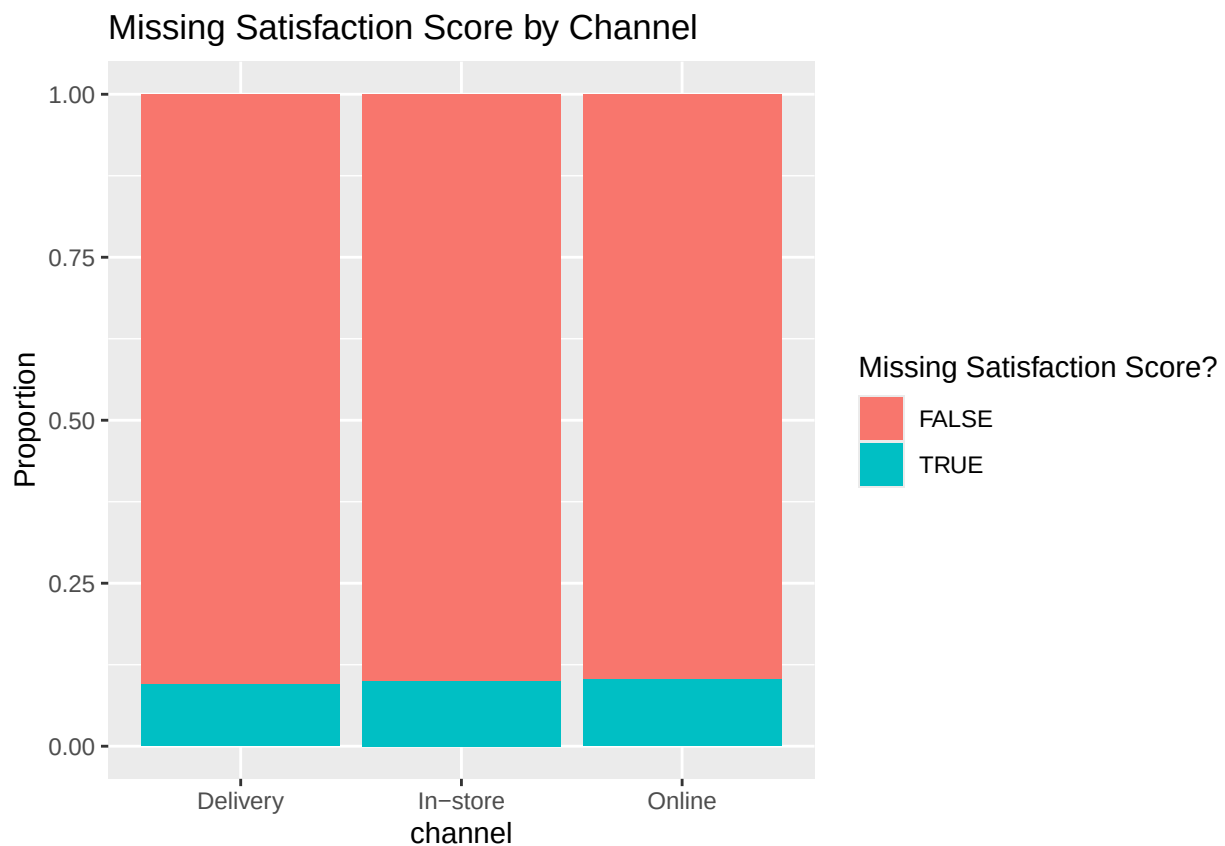
```
data %>%
```



```
group_by(channel) %>%
  summarise(missing_rate = mean(is.na(satisfaction_score)) * 100)
```

```
## # A tibble: 3 x 2
##   channel missing_rate
##   <chr>         <dbl>
## 1 Delivery      9.51
## 2 In-store     10.0
## 3 Online       10.3
```

```
##visuals for satisfaction score
ggplot(data, aes(x = channel, fill = is.na(satisfaction_score))) +
  geom_bar(position = "fill") +
  labs(title = "Missing Satisfaction Score by Channel",
       y = "Proportion",
       fill = "Missing Satisfaction Score?")
```



```
###from the graph we can see that more or less the missingness is similar across all channels
### this may implies a random non response as not all customers are surveyed
```

```
###variable 9: store_lat, store_lon
```

```
missing_coords <- data %>%
  filter(is.na(store_lat) | is.na(store_lon))
nrow(missing_coords)
```

```
## [1] 140
```

```
###which store is affected
```

```
missing_coords %>%  
  count(store_name, district, sort = TRUE)
```

```
##           store_name      district  n  
## 1  IslandMart Store 02           Moka 16  
## 2  IslandMart Store 04    Pamplemousses 14  
## 3  IslandMart Store 12    Plaines Wilhems 14  
## 4  IslandMart Store 07      Black River 13  
## 5  IslandMart Store 08      Savanne 13  
## 6  IslandMart Store 09      Port Louis 13  
## 7  IslandMart Store 01    Pamplemousses 11  
## 8  IslandMart Store 03    Plaines Wilhems 11  
## 9  IslandMart Store 05      Black River 10  
## 10 IslandMart Store 10      Black River 10  
## 11 IslandMart Store 11           Moka 9  
## 12 IslandMart Store 06 Rivière du Rempart 6
```

```
##remove only for spacial
```

```
data <- data %>%  
  filter(!is.na(store_lat) & !is.na(store_lon))
```

```
###change format of date and datetime
```

```
data <- data %>%  
  mutate(date = as.Date(date))
```

```
data <- data %>%  
  mutate(datetime = ymd_hms(datetime))
```

```
##check if it has worked
```

```
class(data$date)
```

```
## [1] "Date"
```

```
class(data$datetime)
```

```
## [1] "POSIXct" "POSIXt"
```

```
#####classification of class
```

```
data$channel<-as.factor(data$channel)  
data$customer_gender<-as.factor(data$customer_gender)  
data$payment_method<-as.factor(data$payment_method)  
data$product_subcategory<-as.factor(data  
$product_subcategory)  
data$customer_segment<-as.factor(data$customer_segment)  
data$age_band<-as.factor(data$age_band)  
data$product_category<-as.factor(data$product_category)  
data$campaign<-as.factor(data$campaign)
```

```
nrow(data)
```

```
## [1] 23511
```

```
###save the MASTERDATASET as cleaned_data
write_csv(data, "cleaned_data.csv")
data<-read_csv("cleaned_data.csv",header=T)

###filtering data to our focus district/category
###focus district=moka
###focus category = electronics

focus_data<-data %>%
  filter(district == "Moka",
         product_category == "Electronics")
View(focus_data)

#Checking if the filter by MOKA and product category has worked
unique(focus_data$district)
```

```
## [1] "Moka"
```

```
unique(focus_data$product_category)
```

```
## [1] "Electronics"
```

```
nrow(focus_data)
```

```
## [1] 423
```

```
colSums(is.na(focus_data)) %>% sort(decreasing = TRUE)
```

##	delivery_minutes	satisfaction_score	transaction_id	datetime
##	258	45	0	0
##	date	year	month	year_month
##	0	0	0	0
##	year_quarter	store_id	store_name	district
##	0	0	0	0
##	store_lat	store_lon	customer_id	customer_gender
##	0	0	0	0
##	age_band	customer_segment	channel	payment_method
##	0	0	0	0
##	campaign	product_category	product_subcategory	quantity
##	0	0	0	0
##	unit_price_mur	discount_pct	gross_margin_pct	gross_sales_mur
##	0	0	0	0
##	net_sales_mur	cost_mur	gross_profit_mur	returned_flag
##	0	0	0	0
##	calc_net_sales	diff		
##	0	0		

```

###class for date/datetime
focus_data <- focus_data %>%
  mutate(date = as.Date(date))

focus_data <- focus_data %>%
  mutate(datetime = ymd_hms(datetime))

###classification of variables
focus_data$channel<-as.factor(focus_data$channel)
focus_data$customer_gender<-as.factor(focus_data$customer_gender)
focus_data$payment_method<-as.factor(focus_data$payment_method)
focus_data$product_subcategory<-as.factor(focus_data
$product_subcategory)
focus_data$customer_segment<-as.factor(focus_data$customer_segment)
focus_data$age_band<-as.factor(focus_data$age_band)
focus_data$product_category<-as.factor(focus_data$product_category)
focus_data$campaign<-as.factor(focus_data$campaign)

```

The initial dataset had 24000 observations; after applying all the cleaning measures we were left with 23511 observations. After filtering for Moka, with Electronics as Product category we have 423 observations to work with.

TASK 2 — UNIVARIATE EDA + VISUAL DESIGN

2.1 Overview

This univariate exploratory data analysis (EDA) examines the statistical characteristics of the dataset focusing on the Moka district and the Electronics product category. The purpose of this analysis is to understand the distribution, variability and statistical behaviour of the variables before conducting further modelling. The dataset contains 423 observations. The main numerical variables analysed include Net Sales, Gross Profit, Delivery Minutes and Satisfaction Score. The categorical variables include Payment Method, Customer Gender, Customer Segment and Channel.

```
str(focus_data)
```

```
## 'data.frame': 423 obs. of 34 variables:
## $ transaction_id : chr "T0015276" "T0020005" "T0013244" "T0019317" ...
## $ datetime : POSIXct, format: "2024-01-02 09:33:02" "2024-01-04 11:44:39" ...
## $ date : Date, format: "2024-01-02" "2024-01-04" ...
## $ year : int 2024 2024 2024 2024 2024 2024 2024 2024 2024 2024 ...
## $ month : int 1 1 1 1 1 1 1 1 1 1 ...
## $ year_month : chr "2024-01" "2024-01" "2024-01" "2024-01" ...
## $ year_quarter : chr "2024Q1" "2024Q1" "2024Q1" "2024Q1" ...
## $ store_id : int 2 11 2 2 11 2 11 2 11 11 ...
## $ store_name : chr "IslandMart Store 02" "IslandMart Store 11" "IslandMart Store 02" "IslandMart Store 11" ...
## $ district : chr "Moka" "Moka" "Moka" "Moka" ...
## $ store_lat : num -20.2 -20.2 -20.2 -20.2 -20.2 ...
## $ store_lon : num 57.5 57.5 57.5 57.5 57.5 ...
## $ customer_id : int 1737 1446 134 1891 2181 294 2357 2075 1933 762 ...
## $ customer_gender : Factor w/ 3 levels "Female","Male",...: 2 1 2 2 1 1 1 1 2 3 ...
## $ age_band : Factor w/ 5 levels "18-24","25-34",...: 1 4 3 5 1 2 2 5 3 1 ...
## $ customer_segment : Factor w/ 3 levels "New","Returning",...: 1 2 2 1 2 2 1 1 1 3 ...
## $ channel : Factor w/ 3 levels "Delivery","In-store",...: 1 1 3 2 2 2 3 2 2 2 ...
## $ payment_method : Factor w/ 4 levels "BankTransfer",...: 2 4 2 3 2 2 2 3 3 2 ...
## $ campaign : Factor w/ 5 levels "BackToSchool",...: 3 3 3 3 3 3 3 3 3 3 ...
## $ product_category : Factor w/ 1 level "Electronics": 1 1 1 1 1 1 1 1 1 1 ...
## $ product_subcategory: Factor w/ 3 levels "Accessories",...: 3 2 1 2 1 3 2 3 2 2 ...
## $ quantity : int 1 1 1 1 1 1 1 1 1 1 ...
## $ unit_price_mur : num 4708 4212 4405 6801 6490 ...
## $ discount_pct : num 0.018 0.04 0.046 0.017 0.02 0.042 0.043 0.031 0.025 0.074 ...
## $ gross_margin_pct : num 0.162 0.147 0.178 0.199 0.201 0.153 0.159 0.227 0.144 0.165 ...
## $ gross_sales_mur : num 4708 4212 4405 6801 6490 ...
## $ net_sales_mur : num 4623 4043 4202 6686 6361 ...
## $ cost_mur : num 3874 3449 3454 5355 5082 ...
## $ gross_profit_mur : num 749 594 748 1330 1278 ...
## $ returned_flag : int 0 1 1 0 0 0 0 0 0 0 ...
## $ delivery_minutes : int 48 63 31 NA NA NA 23 NA NA NA ...
## $ satisfaction_score : num 3.3 2.9 NA 3.9 4.3 3.6 4.8 4.8 3.8 NA ...
## $ calc_net_sales : num 4623 4043 4202 6686 6361 ...
## $ diff : num 0.00032 0.0032 0.00012 0.00006 0.0024 ...
```

The structure output shows the dataset contains both numerical and categorical variables. Missing values are mainly present Delivery Minutes and Satisfaction Score, which likely reflects unrecorded information rather than data errors.

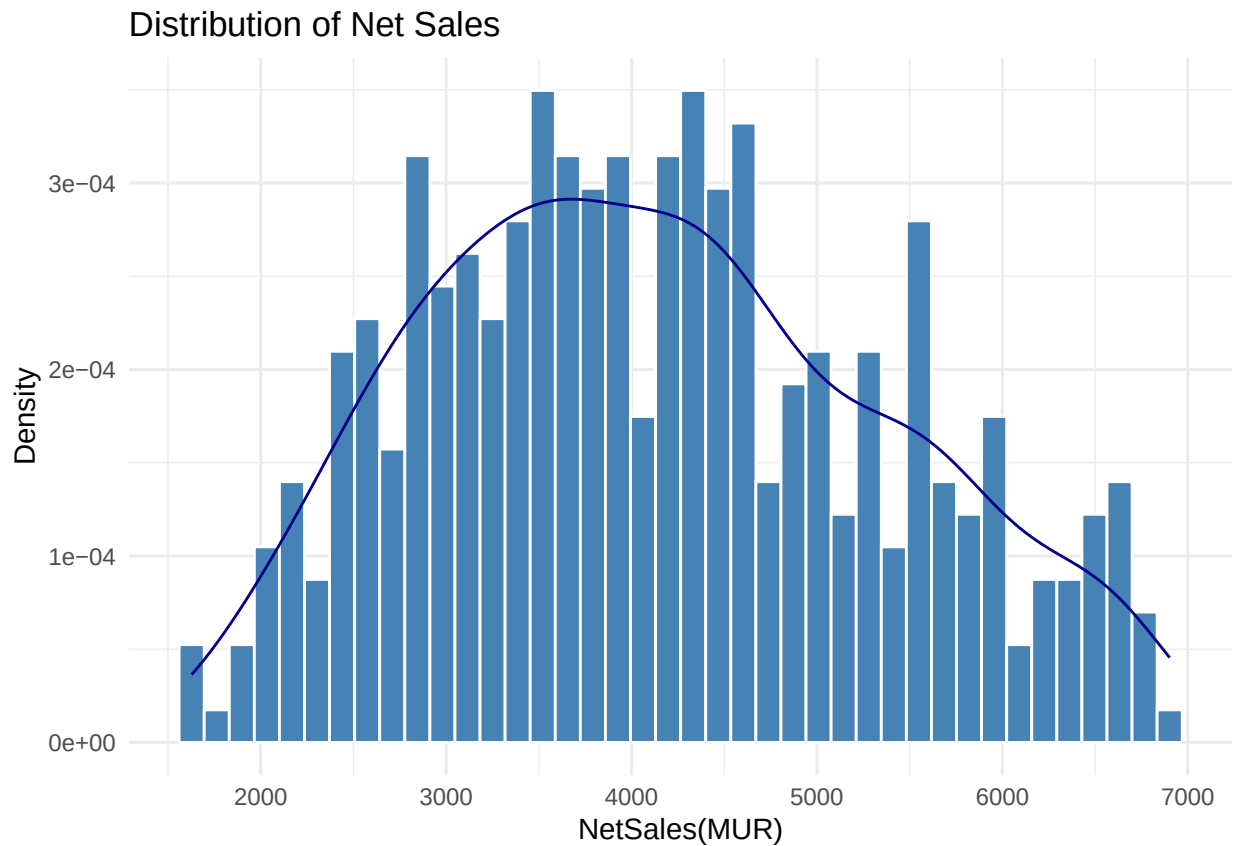
2.2 Numeric Distributions

This section examines the distributional characteristics of the main numeric variables using histograms, density plots, skewness, and kurtosis. The objective is to assess normality, detect skewness, and determine whether transformations are necessary for modelling.

2.2.1 Distribution of Net Sales

A histogram combined with a density plot is used to examine the distribution of Net Sales in order to understand its shape, spread, and possible deviations from normality.

```
ggplot(focus_data,aes(x=net_sales_mur))+  
geom_histogram(aes(y=after_stat(density)),  
bins=40,fill="steelblue",color="white",na.rm=TRUE)+  
geom_density(  
color="darkblue",  
na.rm=TRUE)+  
theme_minimal()+  
labs(title="Distribution of Net Sales",x="NetSales(MUR)",y="Density")
```



```
skewness(focus_data$net_sales_mur,na.rm=TRUE)
```

```
## [1] 0.2513994
```

```
kurtosis(focus_data$net_sales_mur,na.rm=TRUE)
```

```
## [1] 2.291633
```

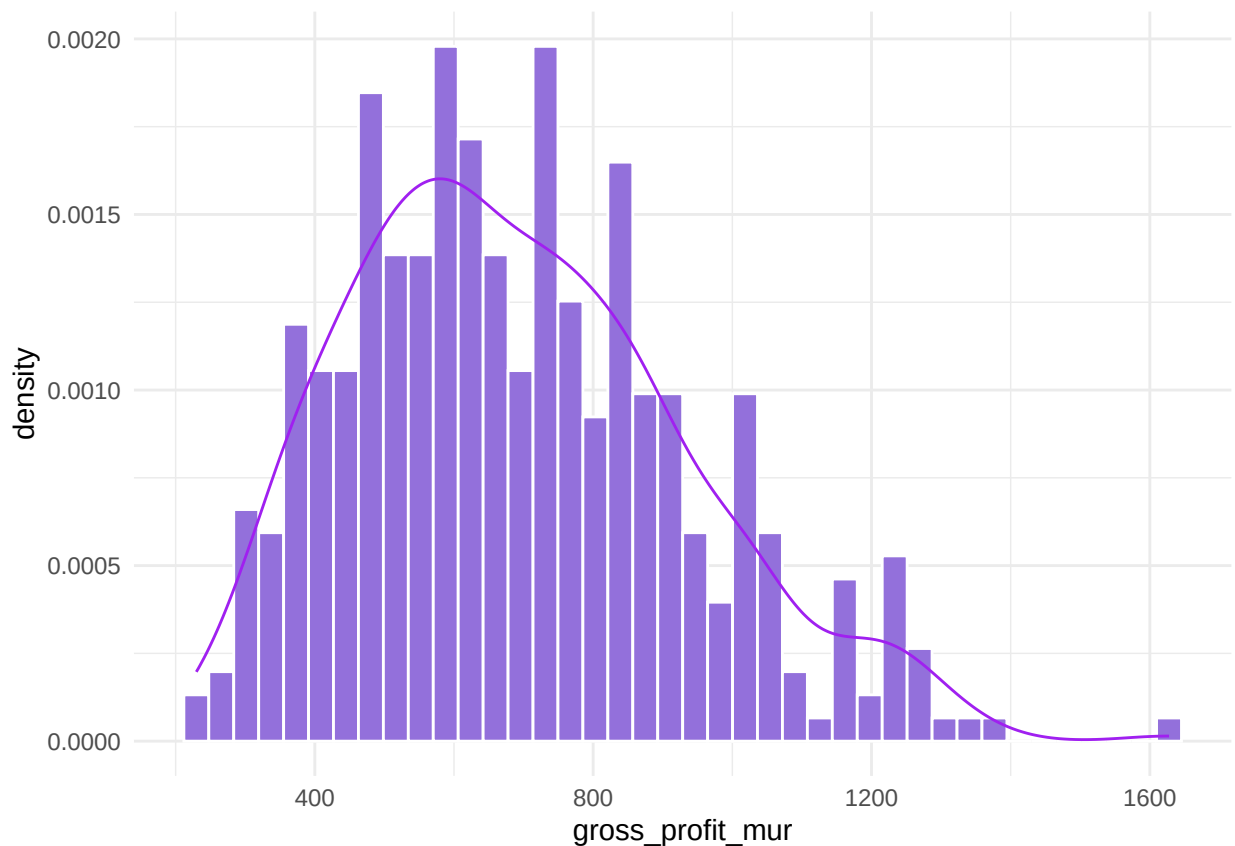
The distribution of Net Sales is approximately symmetric with a slight concentration around the central values. The histogram shows no extreme distortion, and the density curve confirms a fairly balanced shape. Skewness is close to zero, indicating minimal asymmetry in the data. This suggests that sales values are relatively stable across transactions, with no strong dominance of extreme high or low values.

Overall, Net Sales can be considered well-behaved for exploratory analysis, with no need for transformation at this stage.

2.2.2 Distribution of Gross Profit

The distribution of Gross Profit is analysed to identify possible skewness and variability in profit values.

```
ggplot(focus_data,aes(x=gross_profit_mur))+  
geom_histogram(aes(y=after_stat(density)),  
bins=40,fill="mediumpurple",color="white",na.rm=TRUE)+  
geom_density(color="purple",na.rm=TRUE)+  
theme_minimal()
```



```
skewness(focus_data$gross_profit_mur,na.rm=TRUE)
```

```
## [1] 0.5733877
```

```
kurtosis(focus_data$gross_profit_mur,na.rm=TRUE)
```

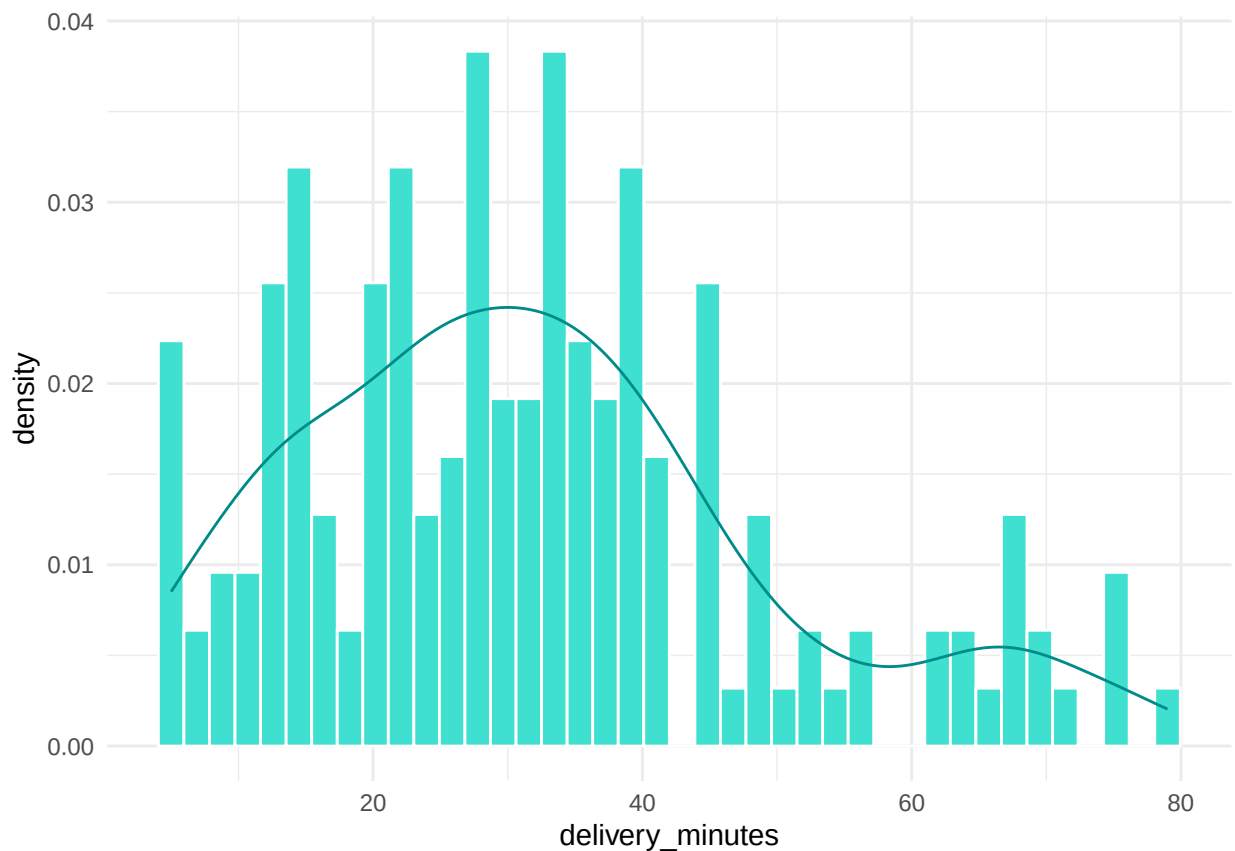
```
## [1] 3.051024
```

Gross Profit shows moderate positive skewness, with most values clustered in the mid-range and a smooth right tail reflecting higher-value transactions. The distribution is unimodal and well-structured, with no significant outliers. Overall, it remains suitable for interpretation in its original scale for univariate analysis.

2.2.3 Delivery Minutes Distribution

Delivery time is analysed to better understand operational efficiency and variability in service performance.

```
ggplot(focus_data,aes(x=delivery_minutes))+  
geom_histogram(aes(y=after_stat(density)),  
bins=40,  
fill="turquoise",  
color="white",  
na.rm=TRUE)+  
geom_density(  
color="darkcyan",  
na.rm=TRUE)+  
theme_minimal()
```



```
cat("Delivery Minutes Skewness:", skewness(focus_data$delivery_minutes, na.rm = TRUE), "\n")
```

```
## Delivery Minutes Skewness: 0.6830888
```

```
cat("Delivery Minutes Kurtosis:", kurtosis(focus_data$delivery_minutes, na.rm = TRUE), "\n")
```

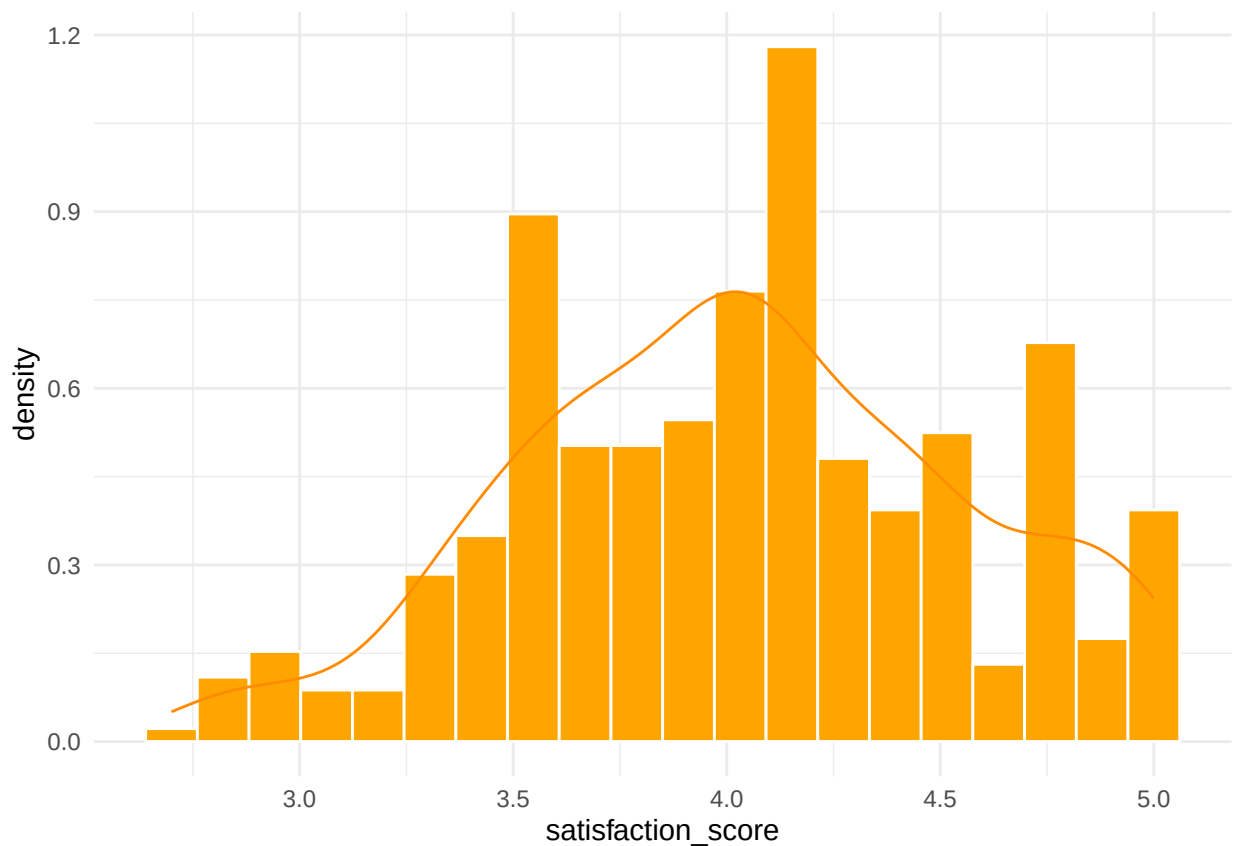
```
## Delivery Minutes Kurtosis: 3.082072
```


The distribution of delivery time is moderately positively skewed, with most values clustered between 20 and 40 minutes and a gradual right tail toward longer durations. However, the skewness is not severe, and the distribution remains smooth and unimodal without extreme outliers. Therefore, a log transformation is not necessary, as the data is sufficiently well-behaved for analysis in its current form.

2.2.4 Satisfaction Score Distribution

Customer Satisfaction Scores are analysed to understand overall feedback patterns and detect possible response bias.

```
ggplot(focus_data,aes(x=satisfaction_score))+  
geom_histogram(aes(y=after_stat(density)),  
bins=20,  
fill="orange",  
color="white",  
na.rm=TRUE)+  
geom_density(  
color="darkorange",  
na.rm=TRUE)+  
theme_minimal()
```



```
cat("Satisfaction Score Skewness:", skewness(focus_data$satisfaction_score, na.rm = TRUE), "\n")
```

```
## Satisfaction Score Skewness: -0.06470355
```

```
cat("Satisfaction Score Kurtosis:", kurtosis(focus_data$satisfaction_score, na.rm = TRUE), "\n")
```

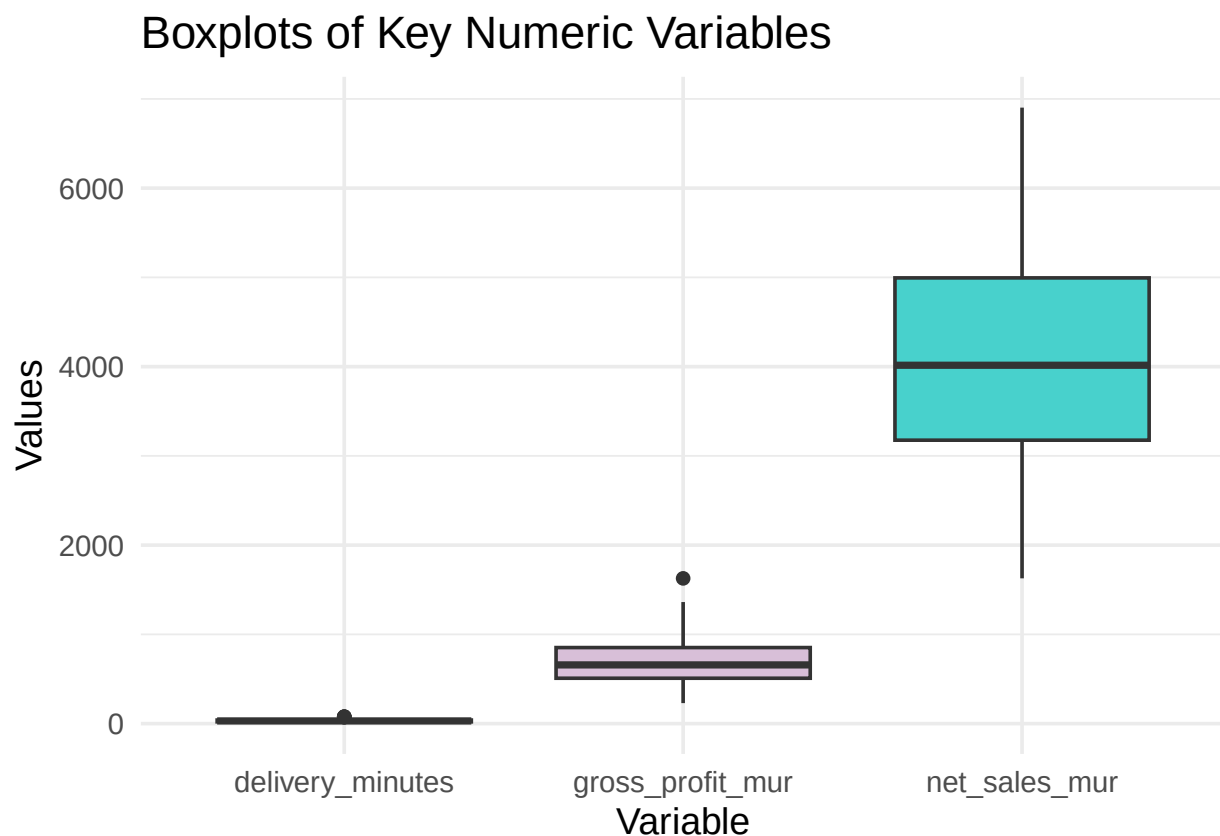
```
## Satisfaction Score Kurtosis: 2.537468
```

Satisfaction Score is nearly perfectly symmetric with no extreme values. The skewness and kurtosis further confirm the absence of extreme response patterns, so no transformation is required.

2.3 Outlier Analysis

Boxplots are used to detect potential outliers.

```
# Pivot data to long format
df_long <- focus_data %>%
  select(net_sales_mur, gross_profit_mur, delivery_minutes) %>%
  pivot_longer(cols = everything(),
    names_to = "Variable",
    values_to = "Value")
# Create combined boxplot
ggplot(df_long, aes(x = Variable, y = Value, fill = Variable)) +
  geom_boxplot(na.rm = TRUE) +
  scale_fill_manual(values = c("lightblue", "thistle", "mediumturquoise")) +
  labs(title = "Boxplots of Key Numeric Variables",
    x = "Variable",
    y = "Values") +
  theme_minimal(base_size = 14) +
  theme(legend.position = "none")
```



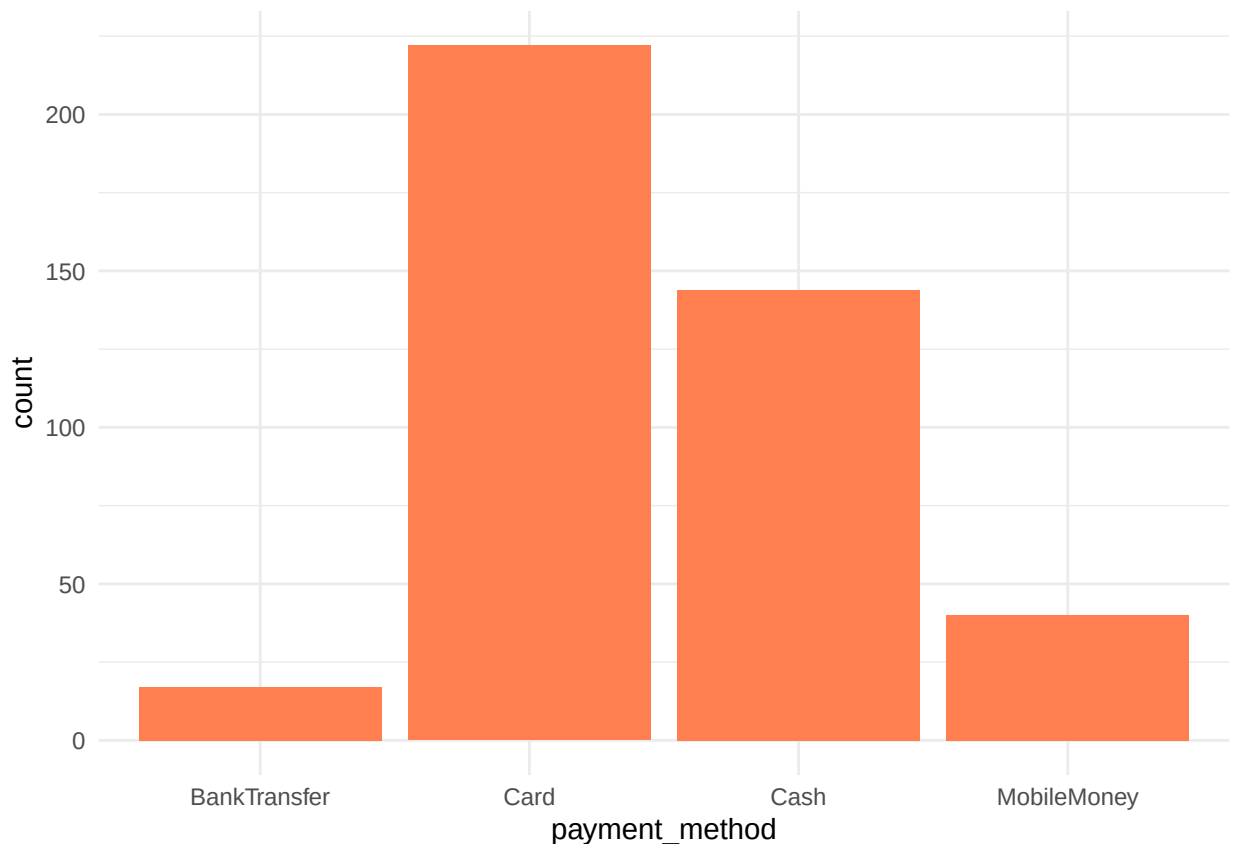
Outliers in the boxplots are identified as individual points beyond the whiskers, which represent values

more than 1.5 times the interquartile range (IQR) from the box. In this case, `delivery_minutes` shows a high outlier above the main distribution, while `gross_profit_mur` also displays an outlier above its upper whisker. In contrast, `net_sales_mur` shows no distinct outlier points, indicating that although it has a wide spread, its values remain within the expected range.

2.4 Categorical Distributions (Bar Charts)

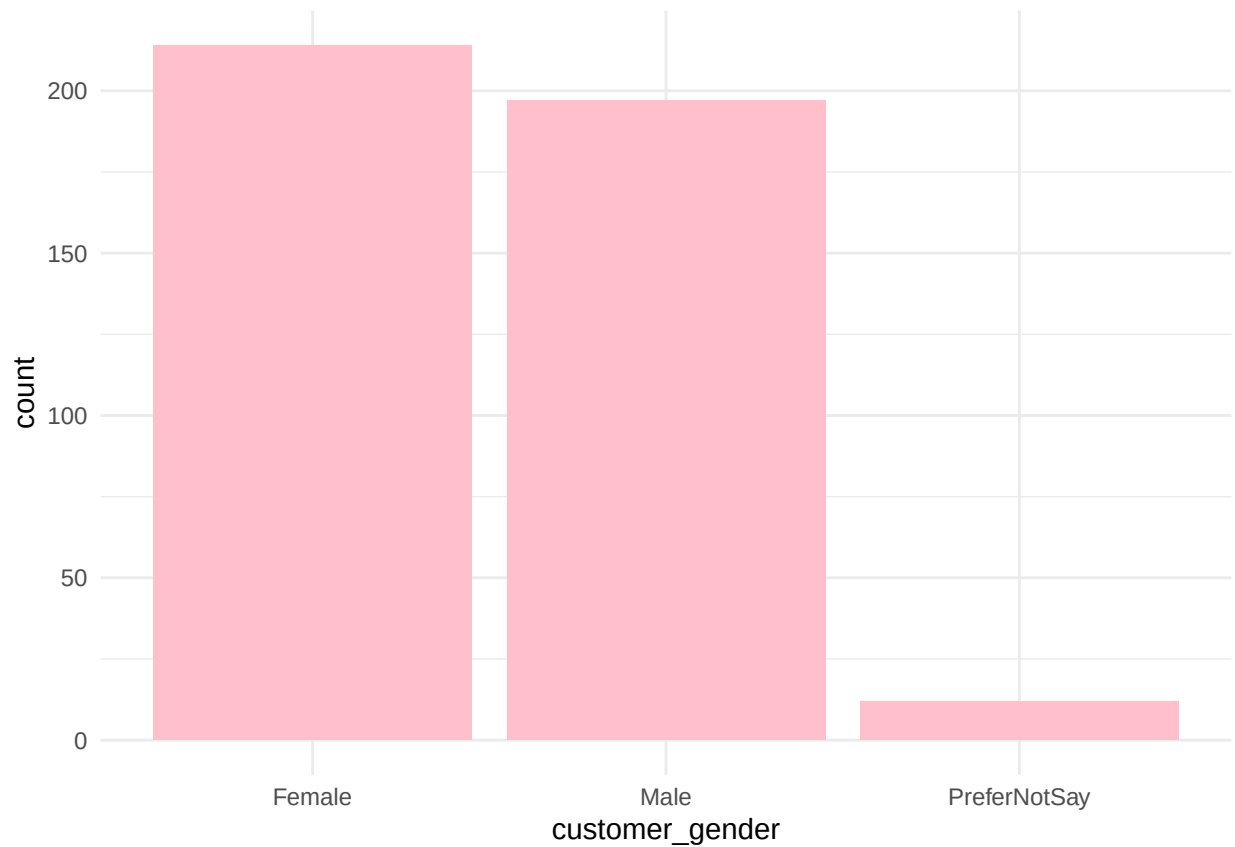
The distribution of key categorical variables is analysed using bar charts to identify dominant patterns and underlying customer behaviour across different dimensions.

```
ggplot(focus_data, aes(x=payment_method)) +  
  geom_bar(fill="coral") +  
  theme_minimal()
```



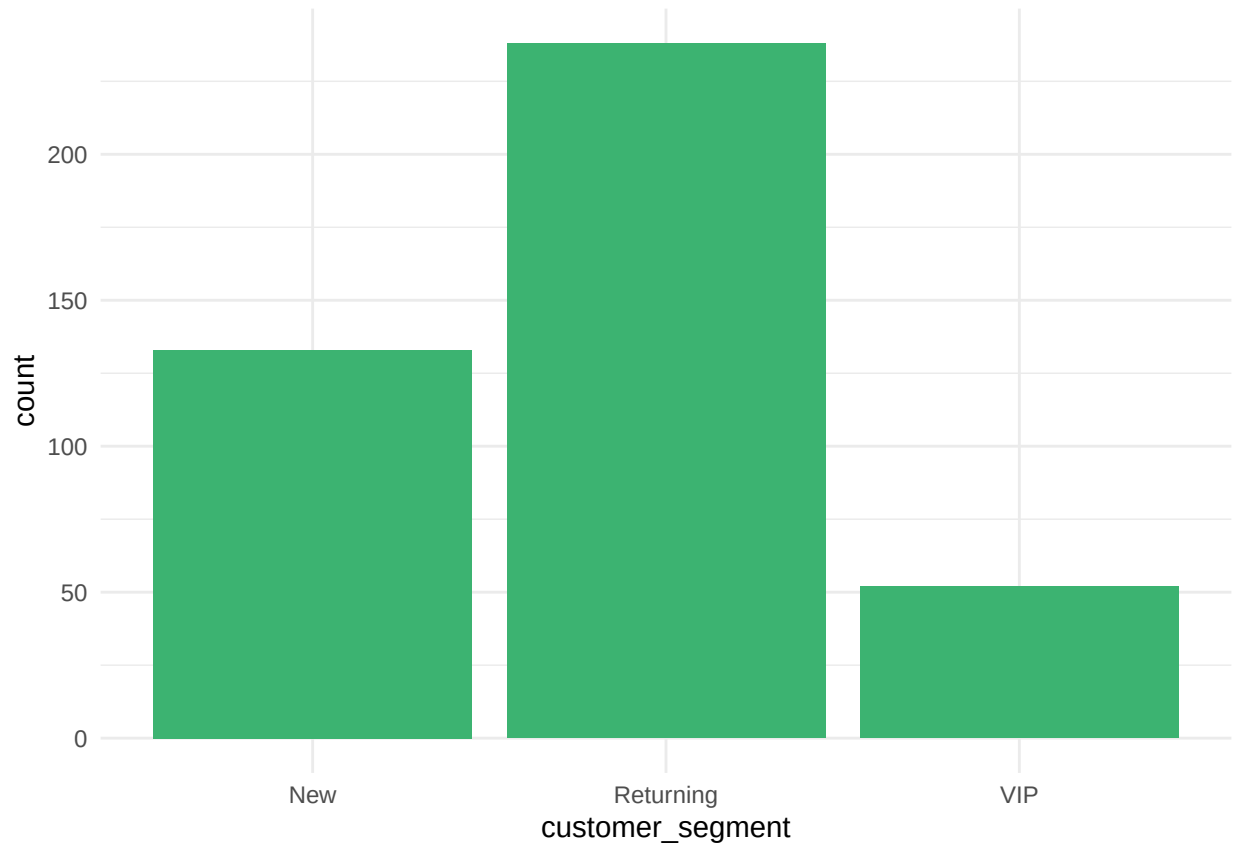
Card payments dominate, indicating strong preference for convenient payment options.

```
ggplot(focus_data, aes(x=customer_gender)) +  
  geom_bar(fill="pink") +  
  theme_minimal()
```



Gender distribution is relatively balanced, with a slight predominance of female customers. This suggests no major gender disparity in purchasing behaviour.

```
ggplot(focus_data, aes(x=customer_segment)) +  
  geom_bar(fill="mediumseagreen") +  
  theme_minimal()
```

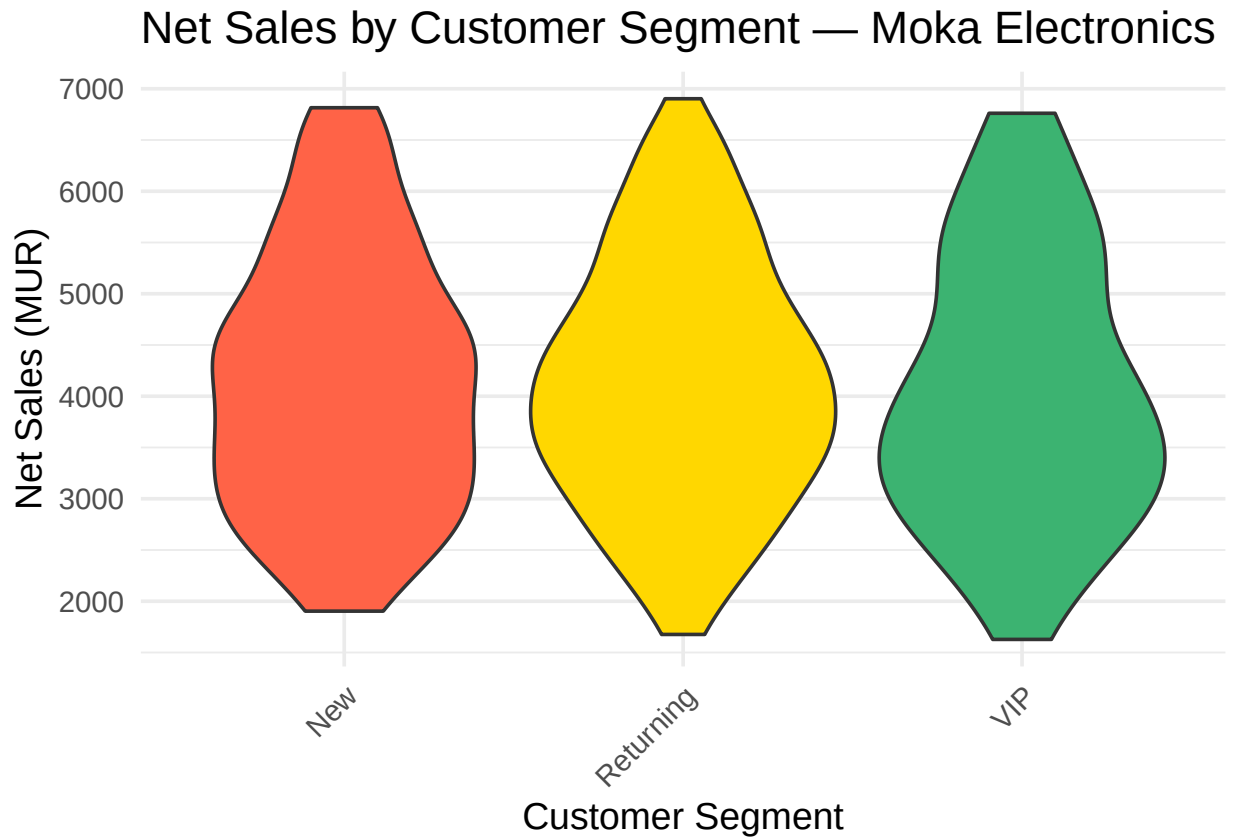


Returning customers form the largest segment, followed by new customers, with VIP customers being the smallest group. This indicates strong customer retention and a stable revenue base.

Grouped Distribution(Violin Plots)

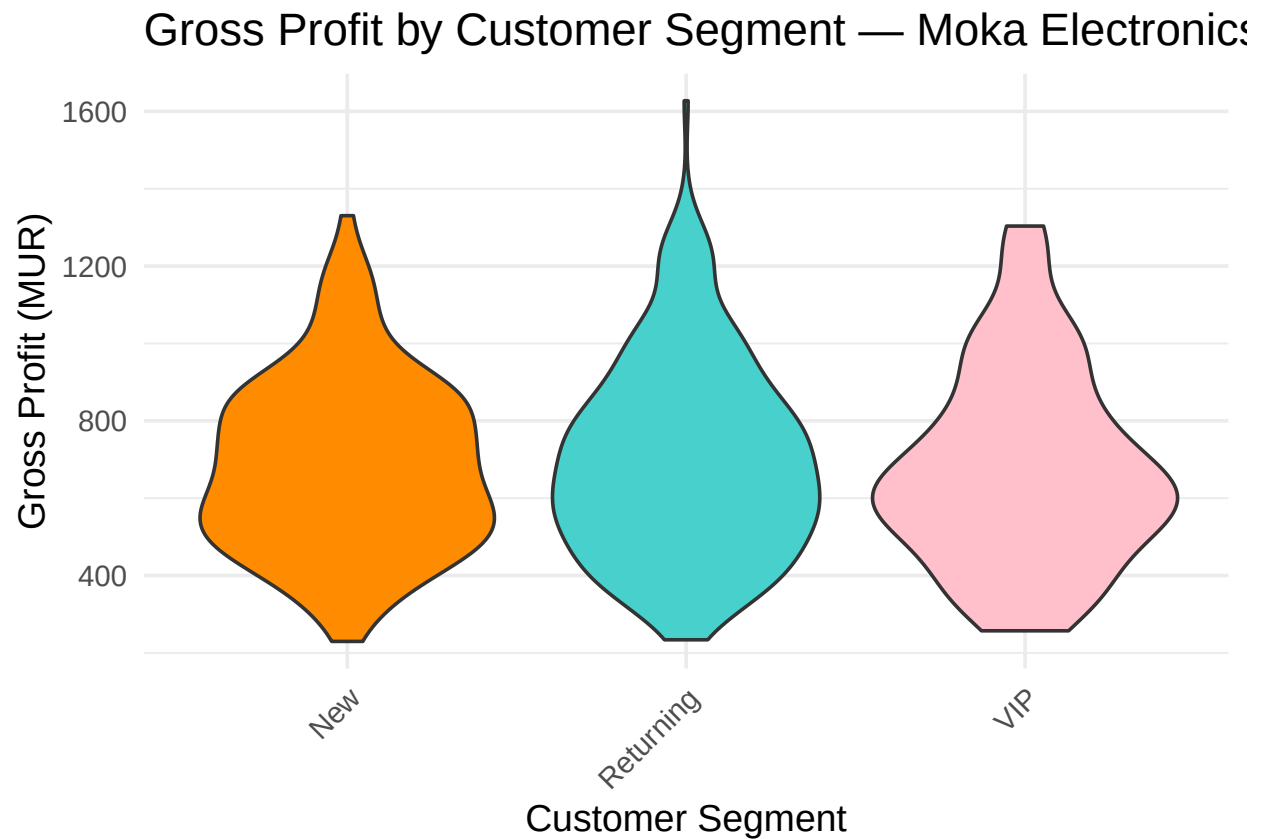
The following violin plots show the distribution of key numeric variables across different groups, highlighting patterns, variability, and segment-specific trends. The violin plots are presented below and can also be found in Appendix 4 of the report.

```
ggplot(focus_data, aes(x = customer_segment, y = net_sales_mur, fill = customer_segment)) +
  geom_violin() +
  labs(title = "Net Sales by Customer Segment - Moka Electronics",
x = "Customer Segment",
y = "Net Sales (MUR)") +
  theme_minimal(base_size = 14) +
  theme(legend.position = "none",
axis.text.x = element_text(angle = 45, hjust = 1)) +
  scale_fill_manual(values = c("tomato","gold","mediumseagreen","skyblue","orchid"))
```



Net Sales are broadly similar across all segments, with Returning customers showing a slightly higher concentration of peak sales around 7,000 MUR.

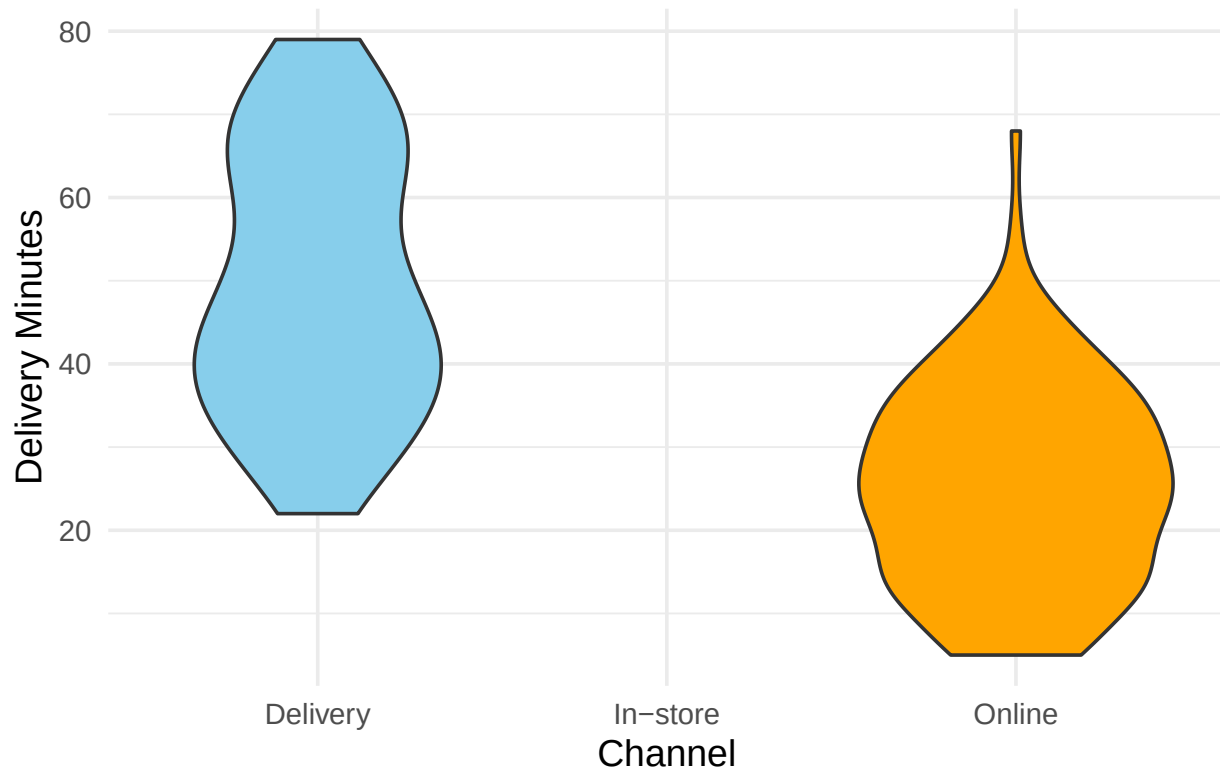
```
ggplot(focus_data, aes(x = customer_segment, y = gross_profit_mur, fill = customer_segment)) +  
  geom_violin() +  
  labs(title = "Gross Profit by Customer Segment - Moka Electronics",  
        x = "Customer Segment",  
        y = "Gross Profit (MUR)") +  
  theme_minimal(base_size = 14) +  
  theme(legend.position = "none",  
        axis.text.x = element_text(angle = 45, hjust = 1)) +  
  scale_fill_manual(values = c("darkorange", "mediumturquoise", "pink", "limegreen", "violet"))
```



Gross Profit is most consistent for VIP customers, while Returning customers exhibit the highest potential upside with a significant “long tail” of high-profit transactions.

```
ggplot(focus_data, aes(x = channel, y = delivery_minutes, fill = channel)) +  
  geom_violin() +  
  labs(title = "Delivery Minutes by Channel - Moka Electronics",  
        x = "Channel",  
        y = "Delivery Minutes") +  
  theme_minimal(base_size = 14) +  
  theme(legend.position = "none") +  
  scale_fill_manual(values = c("skyblue", "orange", "green"))
```

Delivery Minutes by Channel — Moka Electronics



Delivery times vary by channel: Online orders are generally faster and concentrated around 25 minutes, Delivery orders are longer and more variable, while the In-store channel has no recorded delivery minutes, as these values are missing from the dataset.

Overall, Net Sales and Gross Profit differ slightly across customer segments, with Returning customers showing higher peaks and potential profit. Delivery times differ by channel, reflecting operational variability and highlighting areas for potential efficiency improvements.

Summary Statistics Table

The detailed summary statistics table is presented below and can also be found in Appendix 3 of the report.

```
# Ensure required packages are installed and loaded
packages <- c("tidyverse", "moments", "knitr", "kableExtra")
for(p in packages){
  if(!require(p, character.only = TRUE)){
    install.packages(p)
    library(p, character.only = TRUE)
  }
}

# Create summary statistics table
summary_table <- data.frame(
  Variable = c("Net Sales", "Gross Profit", "Delivery Minutes", "Satisfaction Score"),
  Mean = c(
    mean(focus_data$net_sales_mur, na.rm=TRUE),
    mean(focus_data$gross_profit_mur, na.rm=TRUE),
    mean(focus_data$delivery_minutes, na.rm=TRUE),
    mean(focus_data$satisfaction_score, na.rm=TRUE)
  ),
```



```

Median = c(
  median(focus_data$net_sales_mur, na.rm=TRUE),
  median(focus_data$gross_profit_mur, na.rm=TRUE),
  median(focus_data$delivery_minutes, na.rm=TRUE),
  median(focus_data$satisfaction_score, na.rm=TRUE)
),
Min = c(
  min(focus_data$net_sales_mur, na.rm=TRUE),
  min(focus_data$gross_profit_mur, na.rm=TRUE),
  min(focus_data$delivery_minutes, na.rm=TRUE),
  min(focus_data$satisfaction_score, na.rm=TRUE)
),
Max = c(
  max(focus_data$net_sales_mur, na.rm=TRUE),
  max(focus_data$gross_profit_mur, na.rm=TRUE),
  max(focus_data$delivery_minutes, na.rm=TRUE),
  max(focus_data$satisfaction_score, na.rm=TRUE)
),
Skewness = c(
  moments::skewness(focus_data$net_sales_mur, na.rm=TRUE),
  moments::skewness(focus_data$gross_profit_mur, na.rm=TRUE),
  moments::skewness(focus_data$delivery_minutes, na.rm=TRUE),
  moments::skewness(focus_data$satisfaction_score, na.rm=TRUE)
),
Kurtosis = c(
  moments::kurtosis(focus_data$net_sales_mur, na.rm=TRUE),
  moments::kurtosis(focus_data$gross_profit_mur, na.rm=TRUE),
  moments::kurtosis(focus_data$delivery_minutes, na.rm=TRUE),
  moments::kurtosis(focus_data$satisfaction_score, na.rm=TRUE)
)
)
# Display table
knitr::kable(summary_table,
caption = "Summary statistics of numeric variables (including skewness and kurtosis)",
digits = 2) %>%
kableExtra::kable_styling(full_width = FALSE, position = "center")

```

Table 1: Summary statistics of numeric variables (including skewness and kurtosis)

Variable	Mean	Median	Min	Max	Skewness	Kurtosis
Net Sales	4128.25	4015.39	1627.71	6902.32	0.25	2.29
Gross Profit	691.76	657.98	229.70	1627.53	0.57	3.05
Delivery Minutes	31.90	30.00	5.00	79.00	0.68	3.08
Satisfaction Score	4.04	4.00	2.70	5.00	-0.06	2.54

Conclusion

The univariate analysis shows that the dataset is generally well-structured and suitable for further analysis. The numerical variables are mostly well-behaved, with Net Sales and Satisfaction Score showing approximately symmetric distributions, while Gross Profit and Delivery Minutes display mild positive skewness due to higher transaction values and longer delivery times. A small number of outliers are present, mainly in Gross Profit and Delivery Minutes, reflecting normal operational variation. The categorical analysis shows a

strong preference for card payments, a dominant returning customer base, and a relatively balanced gender distribution. Differences in Delivery Minutes across channels highlight variation in operational efficiency. Overall, the dataset is consistent, reliable, and appropriate for further analysis.

Task 3 — Bivariate & Multivariate EDA

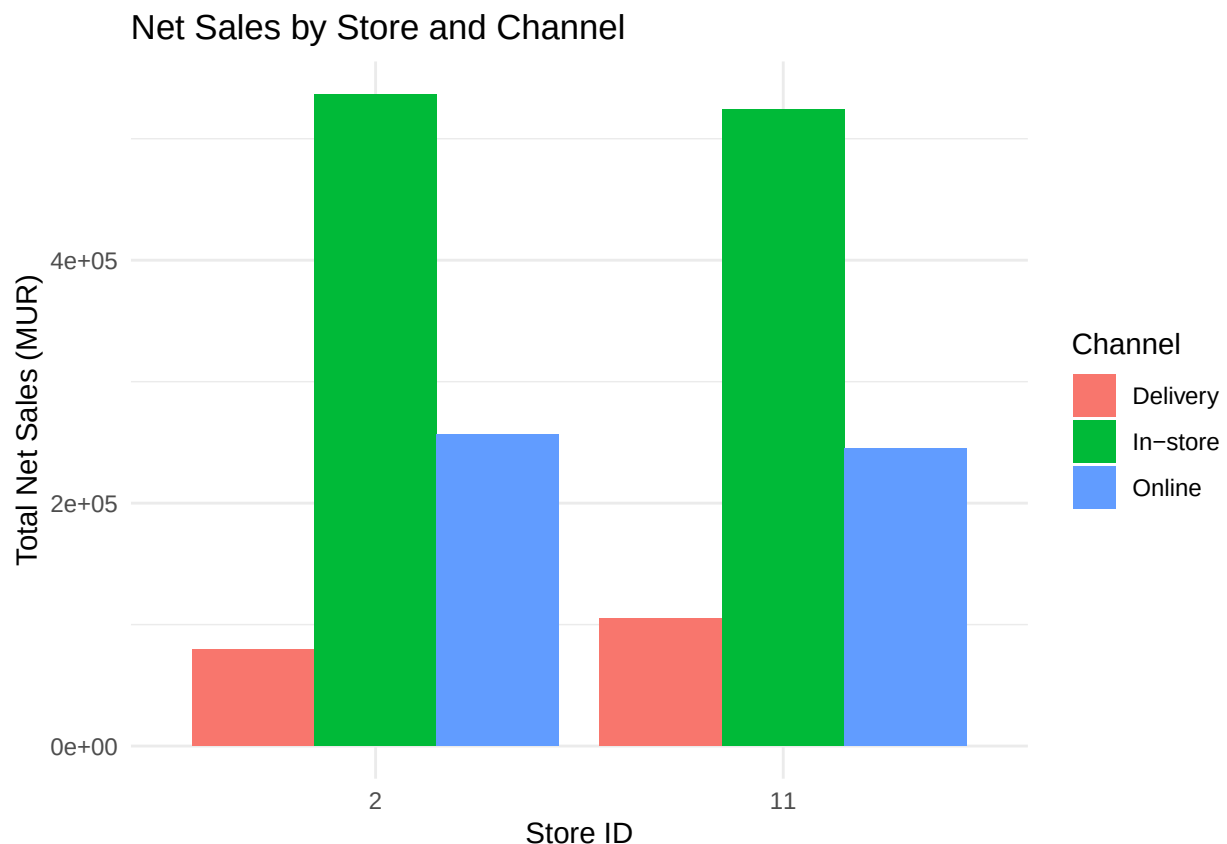
3.1 Overview

This section presents the bivariate and multivariate exploratory data analysis (EDA). The aim is to understand how different variables are related and how they affect business performance and customer behaviour. The analysis focuses on key variables such as net sales, gross profit, customer satisfaction. It also compares results across stores, sales channels, and customer characteristics like gender and age group.

3.2 Net Sales by Store and Channel

The bar chart reveals a significant reliance on physical retail within the IslandMart ecosystem, as in-store sales dramatically outperform both online and delivery channels at Store 2 and Store 11.

```
focus_data %>%
  group_by(store_id, channel) %>%
  summarise(total_net_sales = sum(net_sales_mur, na.rm = TRUE)) %>%
  ggplot(aes(x = factor(store_id), y = total_net_sales, fill = channel)) +
  geom_col(position = "dodge") +
  labs(title = "Net Sales by Store and Channel",
       x = "Store ID",
       y = "Total Net Sales (MUR)",
       fill = "Channel") +
  theme_minimal()
```



3.3 Net Sales by Gender and Store

The bar chart provided offers a detailed demographic breakdown of net sales across IslandMart locations, revealing distinct customer purchasing behaviors between Store 2 and Store 11.

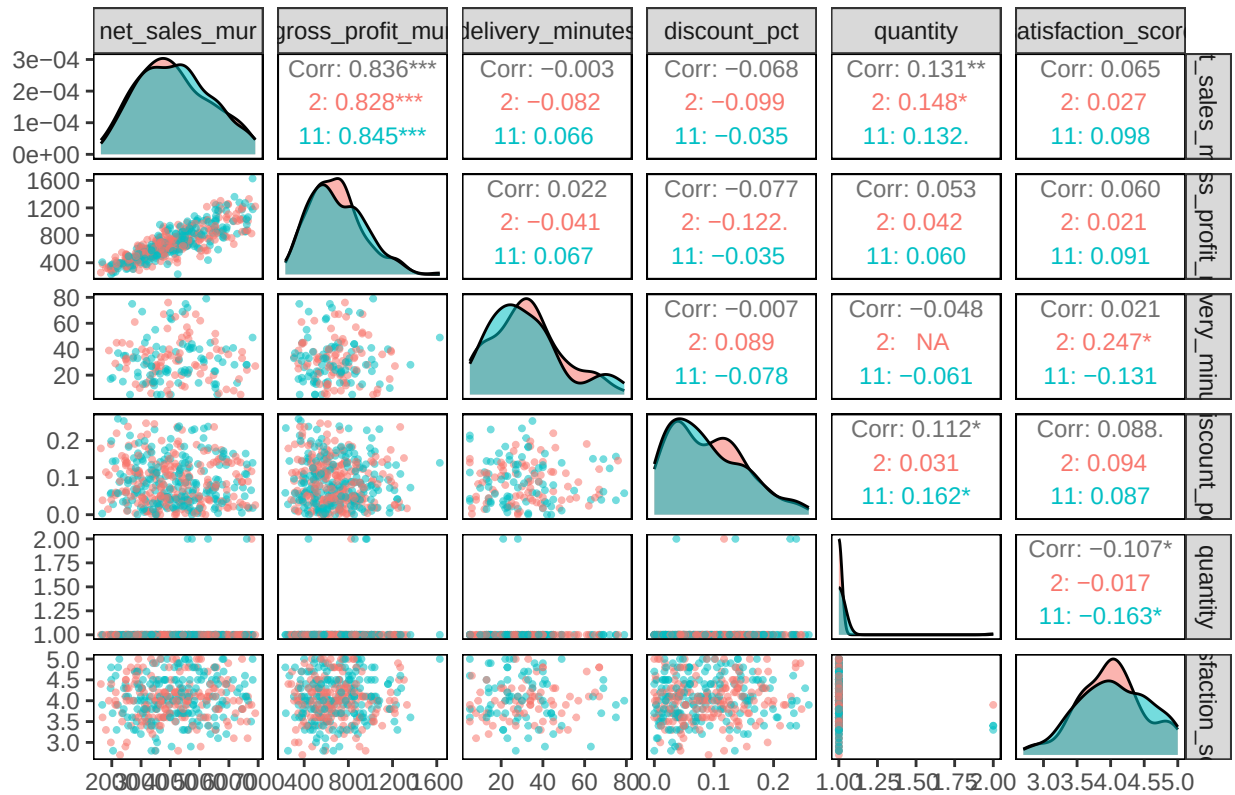
```
focus_data %>%
  group_by(store_id, customer_gender) %>%
  summarise(total_net_sales = sum(net_sales_mur, na.rm = TRUE)) %>%
  ggplot(aes(x = factor(store_id), y = total_net_sales, fill = customer_gender)) +
  geom_col(position = "dodge") +
  labs(title = "Net Sales by Gender and Store",
       x = "Store ID",
       y = "Total Net Sales (MUR)",
       fill = "Gender") +
  theme_minimal()
```



3.4 Multivariate Analysis (Pair Plot)

This correlation matrix provides a sophisticated multivariate analysis of IslandMart's operational health, specifically isolating the drivers of profitability and customer experience for Store 2 and Store 11.

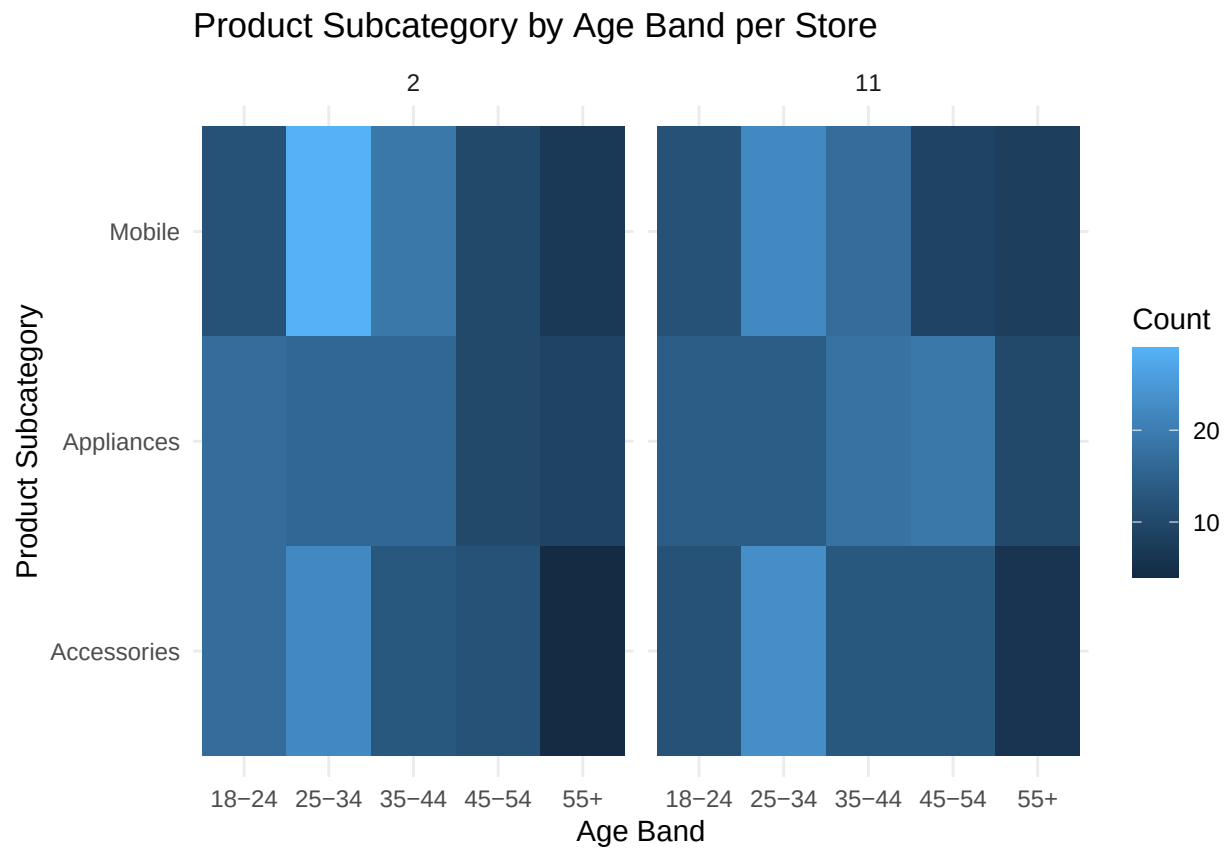
Multivariate Relationships Between Key Variables



3.5 Product Subcategory by Age Band per Store

This heat map provides a multivariate view of IslandMart's customer demographics, crossing product subcategories with age bands across Store 2 and Store 11.

```
focus_data %>%
  count(store_id, age_band, product_subcategory) %>%
  ggplot(aes(x = age_band, y = product_subcategory, fill = n)) +
  geom_tile() +
  facet_wrap(~ store_id) +
  labs(title = "Product Subcategory by Age Band per Store",
       x = "Age Band",
       y = "Product Subcategory",
       fill = "Count") +
  theme_minimal()
```



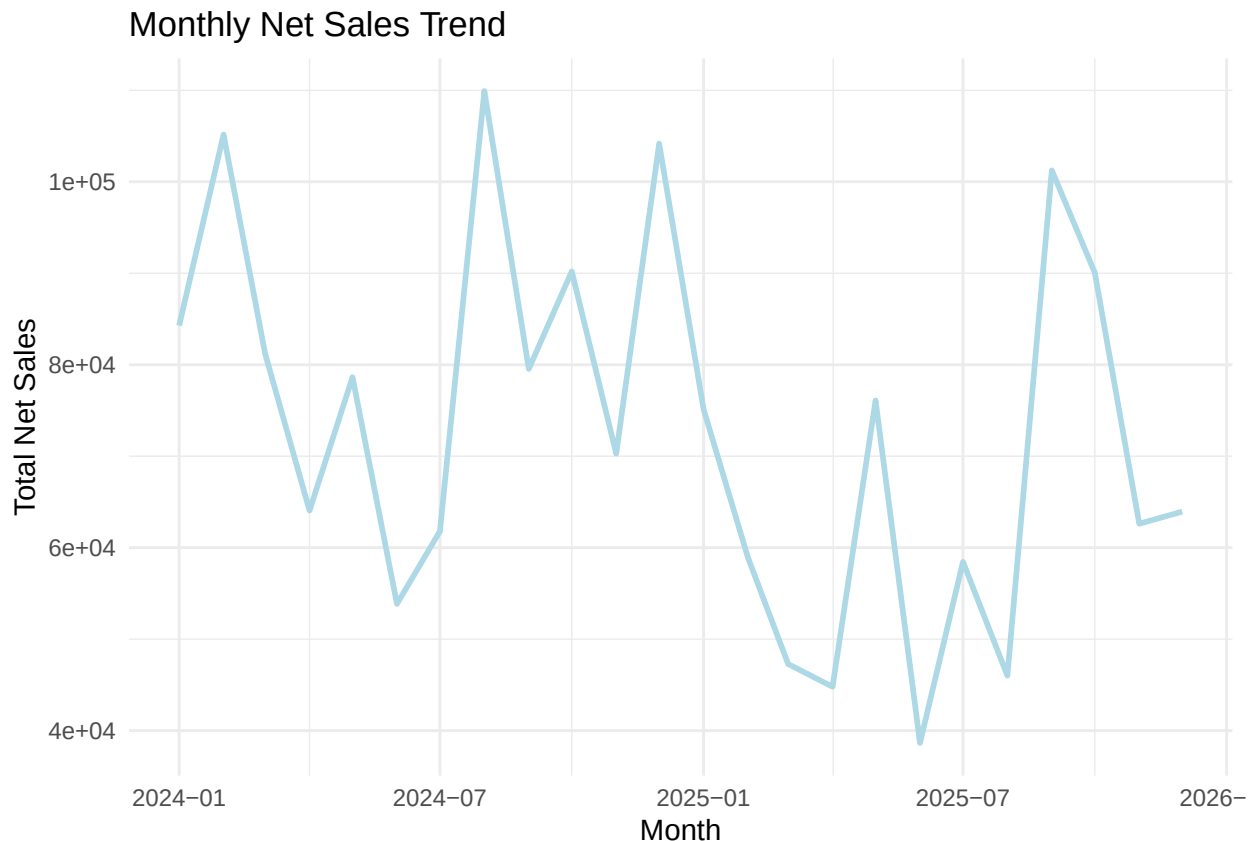
Task 4 : Time-series + special visualization

Under this section, we will look at the sales and profit trends of Electronics of Moka District.

4.1 Time series Analysis

We have two graphs showing the monthly net sales and net sales vs profit respectively.

```
monthly_sales <- focus_data %>%  
  mutate(month = floor_date(date, "month")) %>%  
  group_by(month) %>%  
  summarise(  
    total_sales = sum(net_sales_mur, na.rm = TRUE),  
    total_profit = sum(gross_profit_mur, na.rm = TRUE)  
  )  
ggplot(monthly_sales, aes(x = month, y = total_sales)) +  
  geom_line(color = "lightblue", linewidth=1) +  
  labs(title = "Monthly Net Sales Trend",  
       x = "Month",  
       y = "Total Net Sales") +  
  theme_minimal()
```



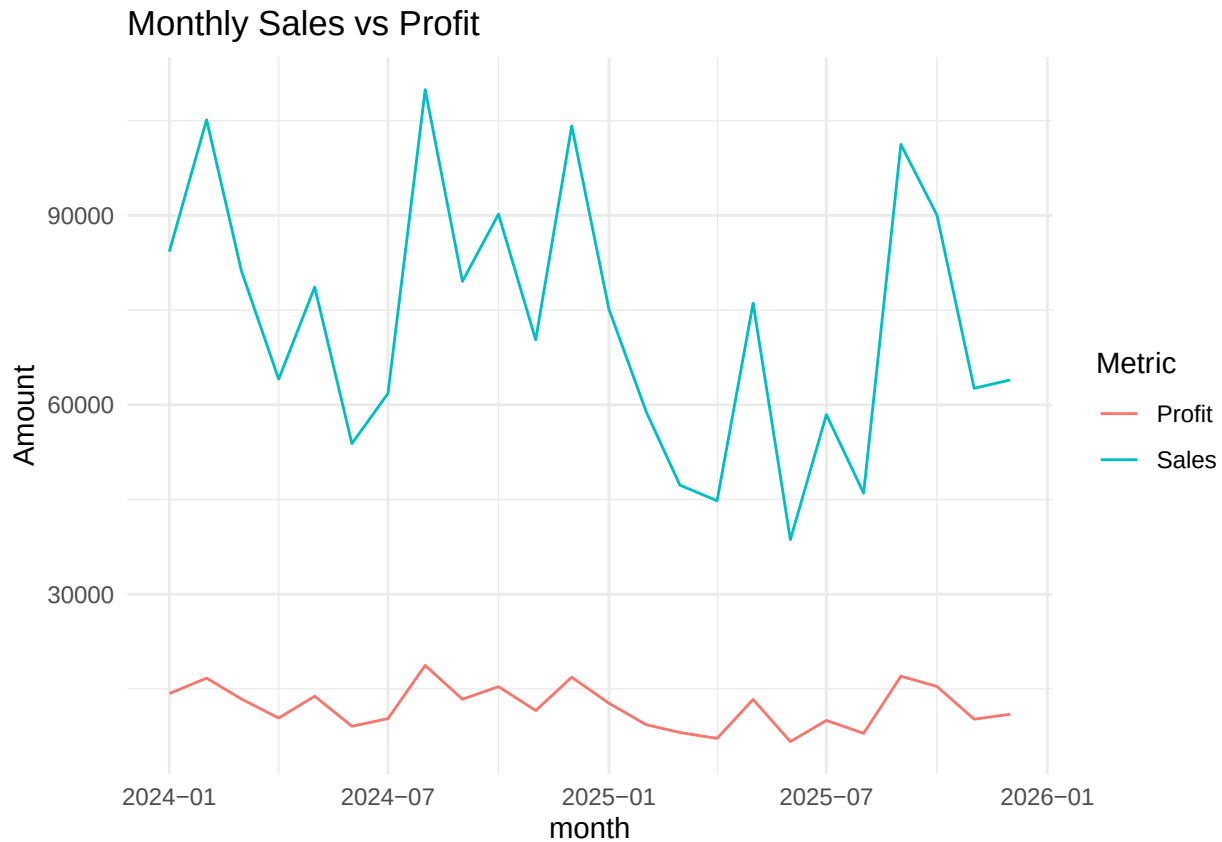
The monthly sales trend shows whether Electronics sales in Moka are stable, growing or fluctuating over time.

```
ggplot(monthly_sales, aes(month)) +  
  geom_line(aes(y = total_sales, color = "Sales")) +  
  geom_line(aes(y = total_profit, color = "Profit")) +  
  labs(title = "Monthly Sales vs Profit",
```

```

y = "Amount",
color = "Metric") +
theme_minimal()

```



Comparing sales and profit helps determine whether higher sales are consistently translating into higher profitability.

4.2 Store Performance and Spatial Analysis

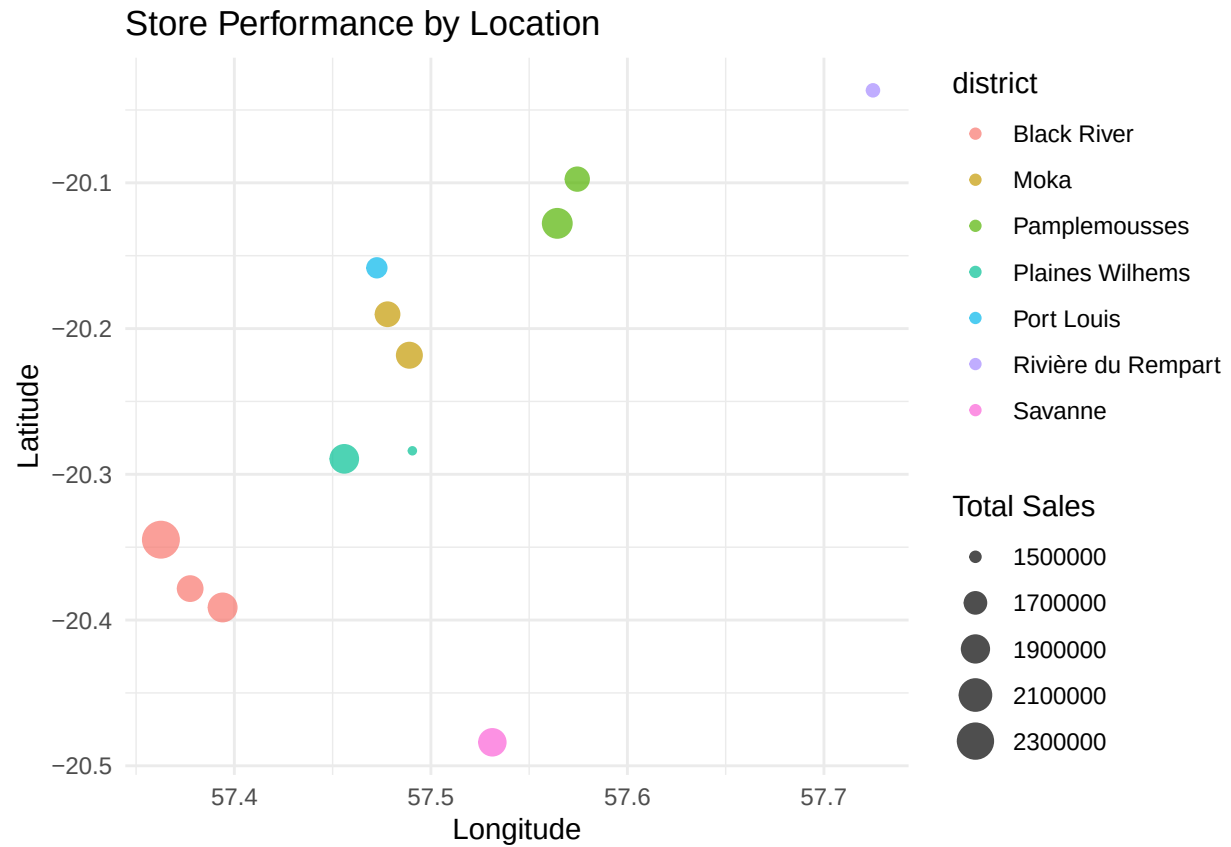
To compare Moka with the wide store network, store-level performance is illustrated below.

```

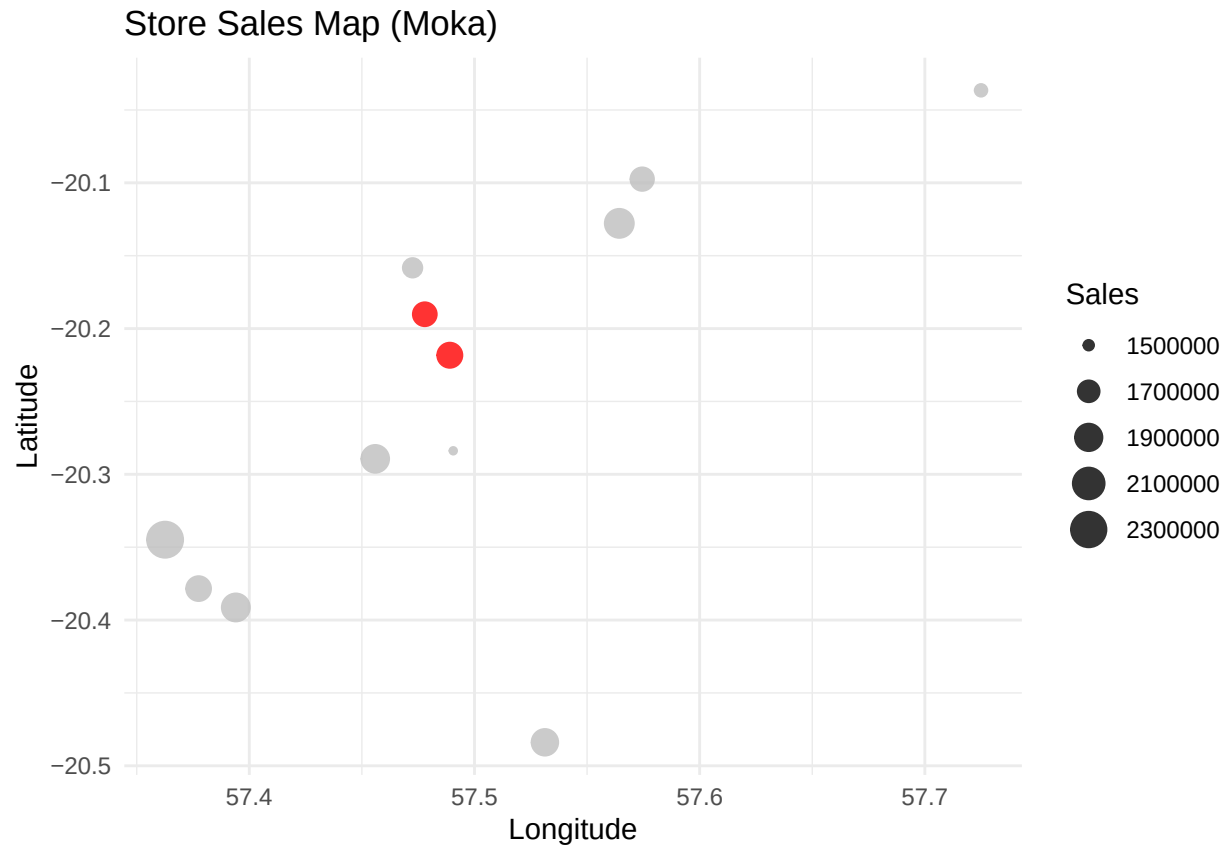
store_map <- data %>%
  group_by(store_name, district, store_lat, store_lon) %>%
  summarise(total_sales = sum(net_sales_mur),
            total_profit = sum(gross_profit_mur),
            .groups = "drop") %>%
  filter(!is.na(store_lat), !is.na(store_lon))

ggplot(store_map, aes(x = store_lon, y = store_lat)) +
  geom_point(aes(size = total_sales, color = district), alpha = 0.7) +
  labs(title = "Store Performance by Location",
       x = "Longitude",
       y = "Latitude",
       size = "Total Sales") +
  theme_minimal()

```

```
ggplot(store_map, aes(store_lon, store_lat)) +
  geom_point(aes(size = total_sales,
                  color = district == "Moka"), alpha = 0.8) +
  scale_color_manual(values = c("TRUE" = "red", "FALSE" = "grey"),
                     guide = "none") +
  labs(title = "Store Sales Map (Moka)",
       x = "Longitude",
       y = "Latitude",
       size = "Sales") +
  theme_minimal()
```



The Moka highlighted map shows where Moka sits spatially relative to the broader store network.

The following table and chart are found in **Appendix 2**

Appendix proof- Districts and No.of stores

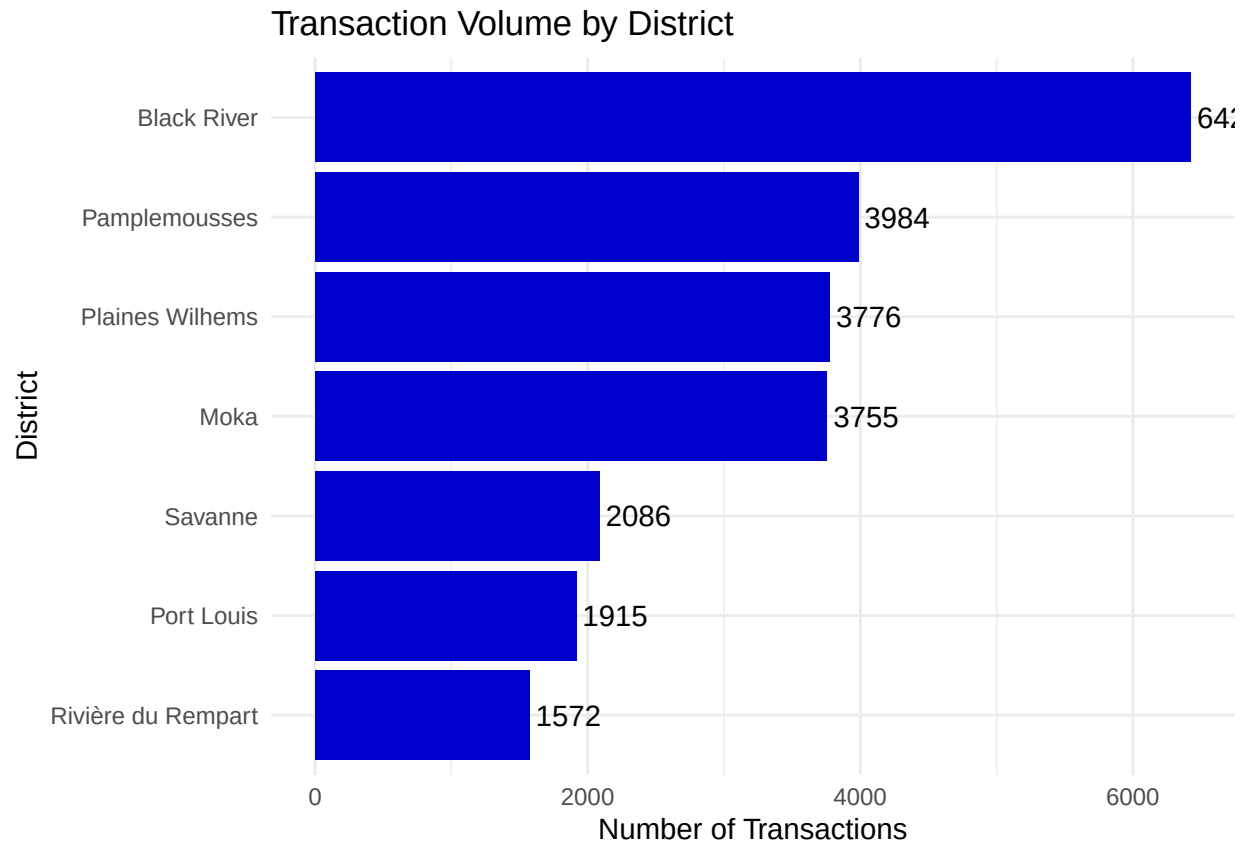
```
data_summary <- data %>%
  group_by(district) %>%
  summarise(
    number_of_stores = n_distinct(store_name),
    number_of_transactions = n()
  ) %>%
  arrange(desc(number_of_transactions))

print(data_summary)
```

```
## # A tibble: 7 x 3
##   district      number_of_stores number_of_transactions
##   <chr>          <int>          <int>
## 1 Black River      3            6423
## 2 Pamplemousses    2            3984
## 3 Plaines Wilhems  2            3776
## 4 Moka             2            3755
## 5 Savanne          1            2086
## 6 Port Louis       1            1915
## 7 Rivière du Rempart 1            1572
```

Transactions count by Districts

```
data %>%
  count(district) %>%
  ggplot(aes(x = reorder(district, n), y = n)) +
  geom_bar(stat = "identity", fill = "mediumblue") +
  geom_text(aes(label = n), hjust = -0.1) +
  coord_flip() +
  labs(title = "Transaction Volume by District",
       x = "District",
       y = "Number of Transactions") +
  theme_minimal()
```



Moka Monthly Transactions

```
focus_data %>%
  count(month) %>%
  mutate(month_name = month.name[month]) %>%
  ggplot(aes(x = factor(month_name, levels = month.name), y = n)) +
  geom_bar(stat = "identity", fill = "lightblue") +
  geom_text(aes(label = n), vjust = -0.3, size = 4) +
  labs(title = "Number of Transactions by Month (Moka - Electronics)",
       x = "Month",
       y = "Transactions") +
  theme_minimal()
```

